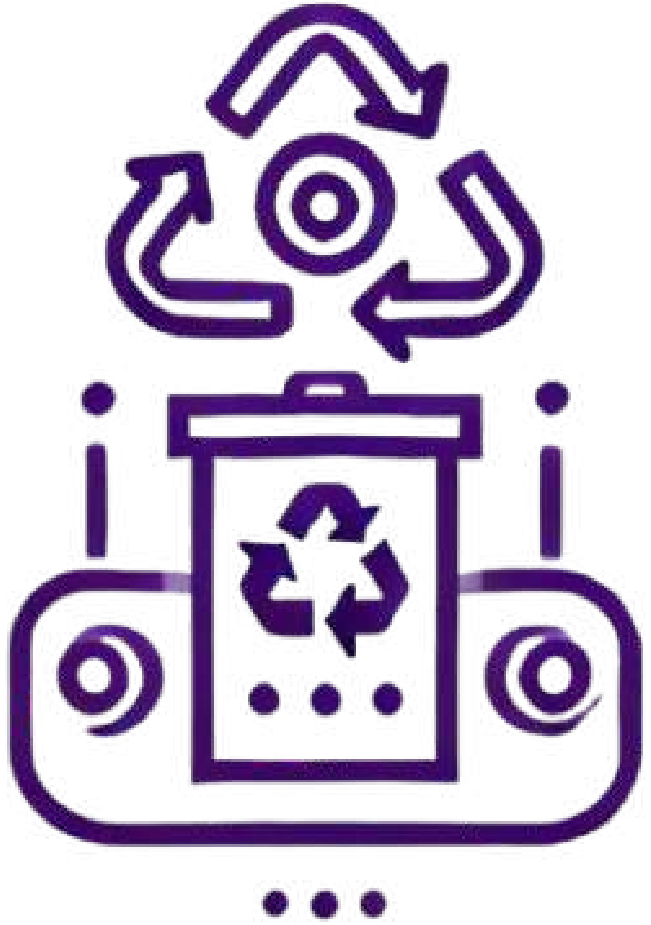
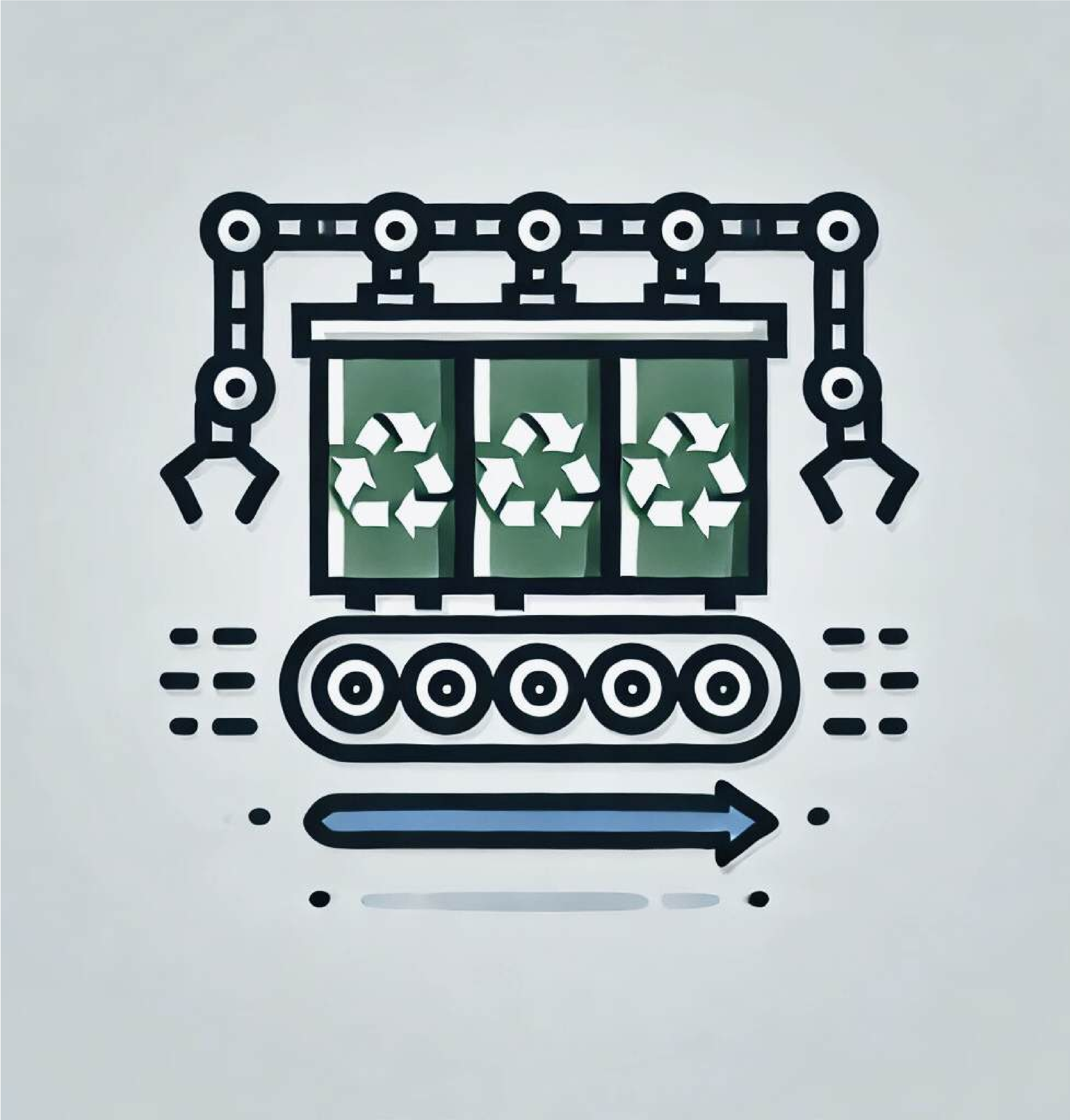


AUTOMATIC
WASTE SEGREGATION



About Automatic Waste Segregation

About Technology

Automated methods of waste disposal have been developed in recent years and several products have been designed to work through mechanisms of automated waste segregation with certain characteristics. These products include AI, robotics, machine learning among others to improve the speed and effectiveness of sorting out the wastes. Besides, the usage of these technologies in the formation of the products serves to minimize time and effort while enhancing the efficiency of environmental control.

Problem Statement

Public places need an efficient and effective way to manage and segregate waste because improper waste segregation in high-traffic areas leads to increased environmental pollution, decreased recycling efficiency, and higher waste management costs. Manual sorting of waste in public spaces is often labour-intensive, prone to errors, and can be unhygienic for workers. An automated waste segregation system designed for public use can address these issues by accurately sorting recyclable materials like metals, plastics, glass, and paper, thereby promoting sustainable waste management practices and reducing the operational burden on public maintenance services.

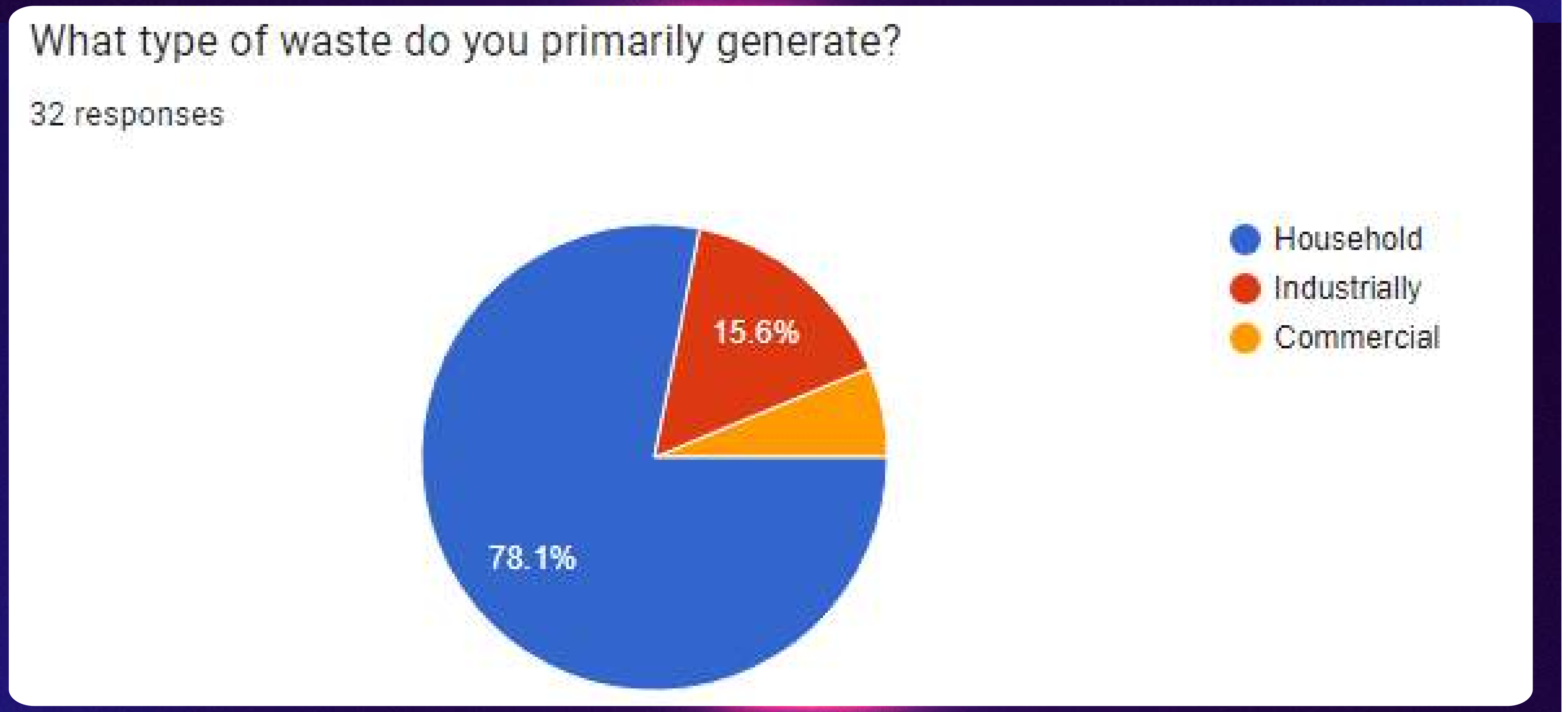
Objectives

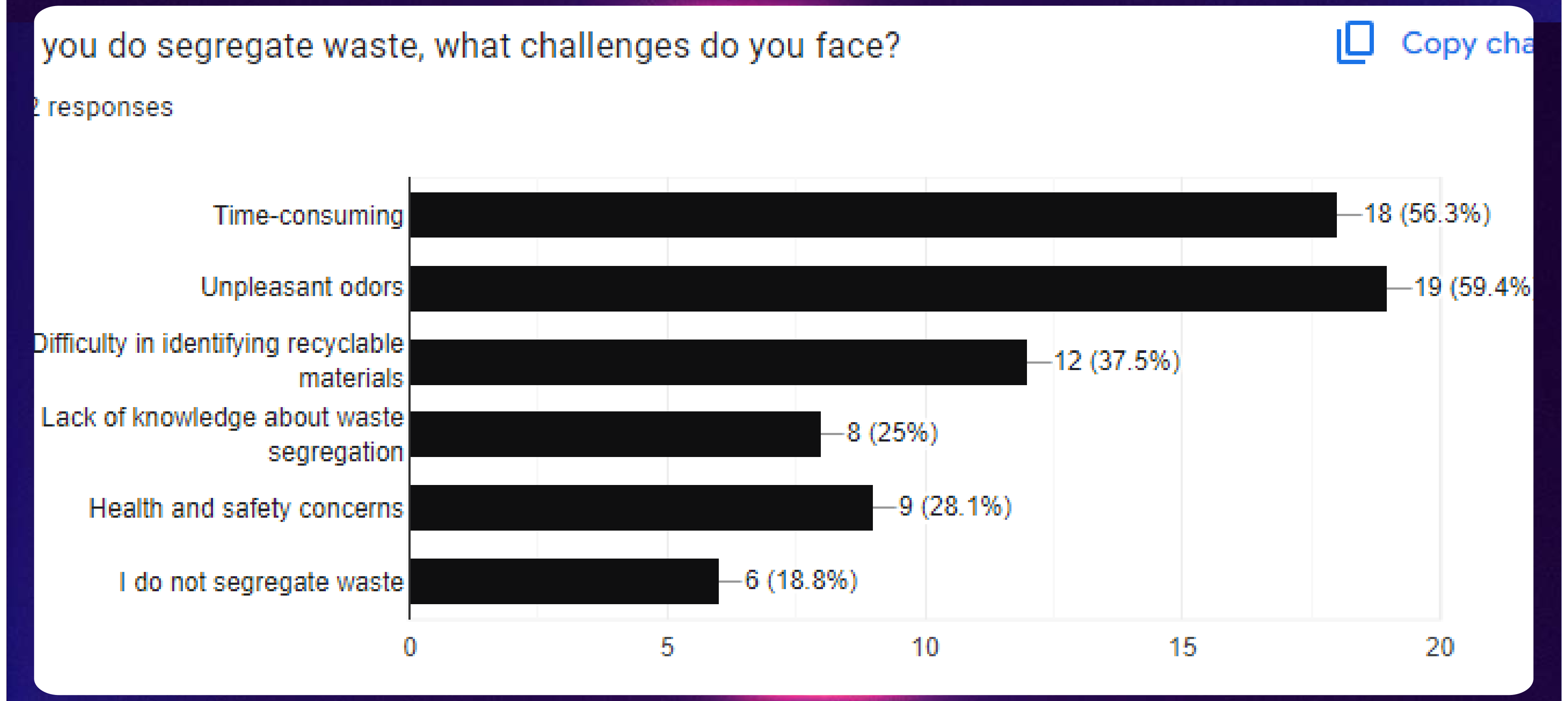
01 To design a machine that can segregate recycled waste (Paper, glass, plastic, and metal) to minimize the manual efforts of sorting.

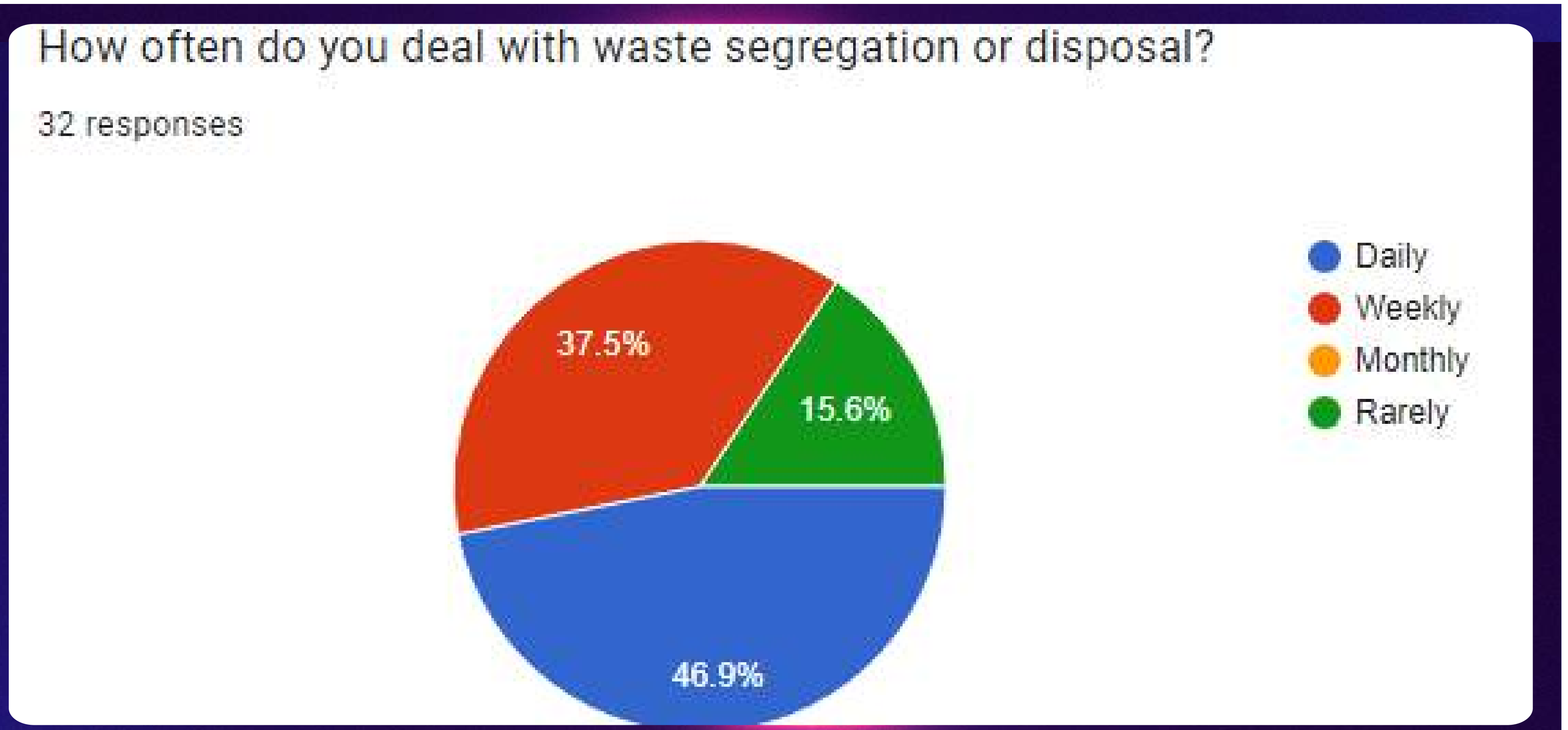
02

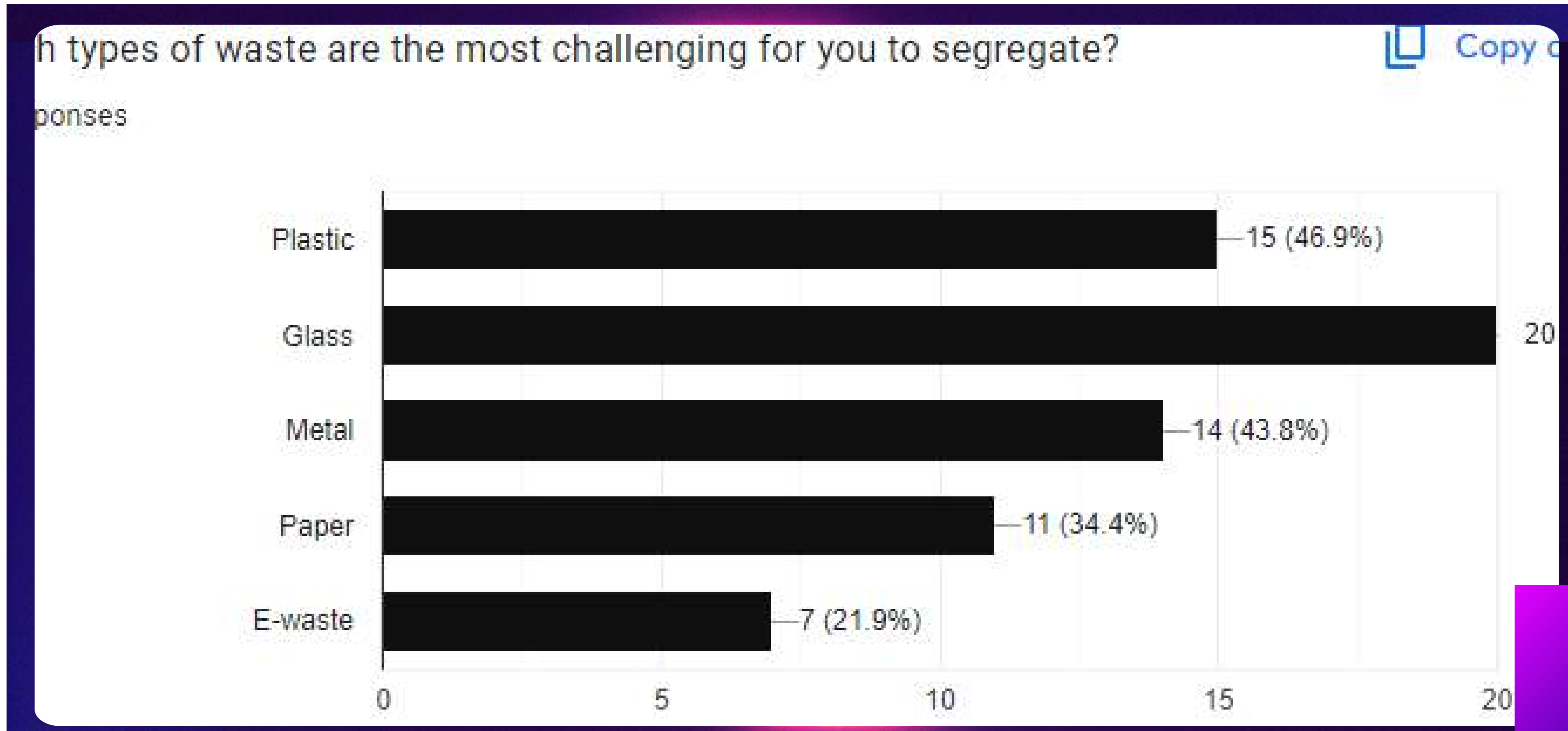
02 To evaluate performance of the machine

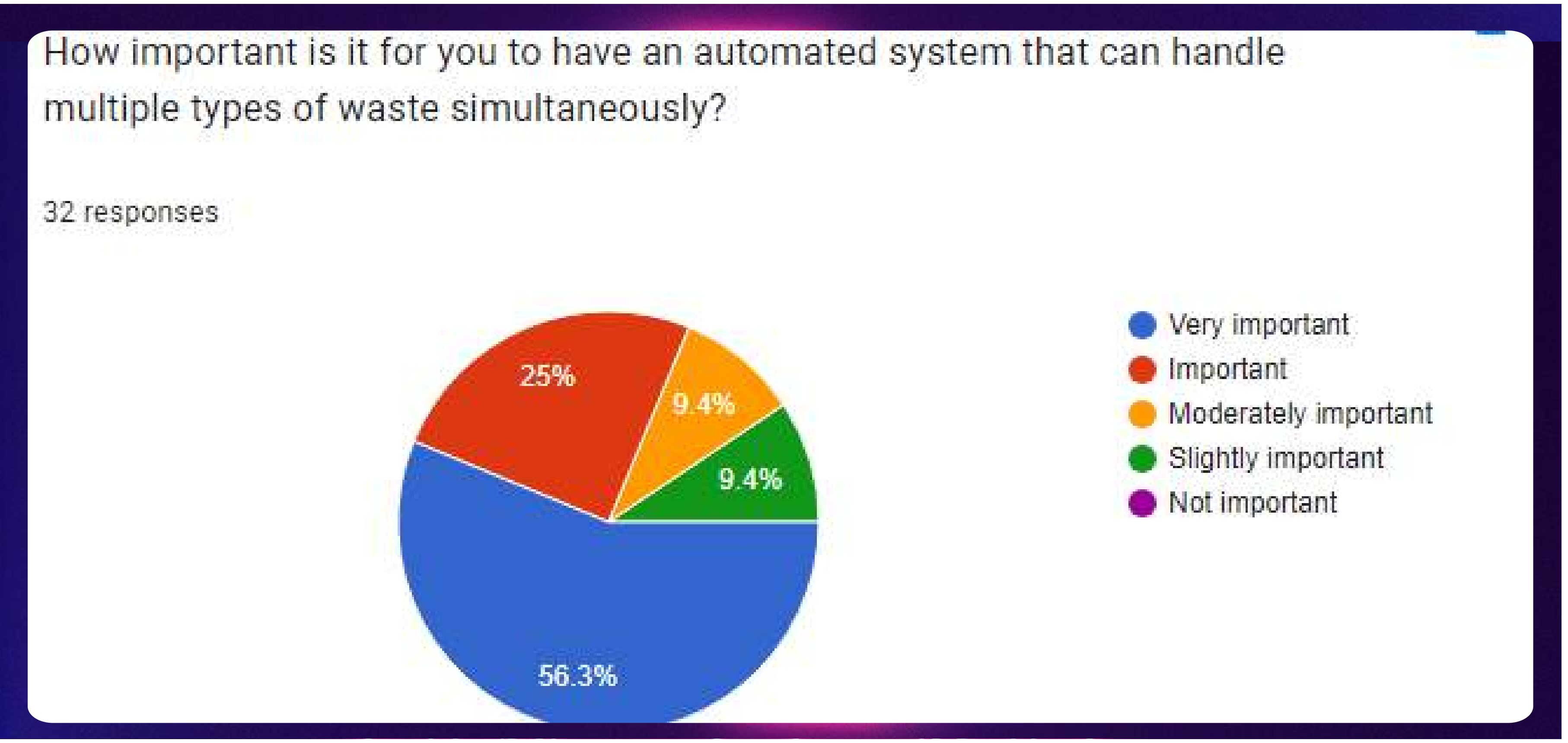
CUSTOMER SURVEY

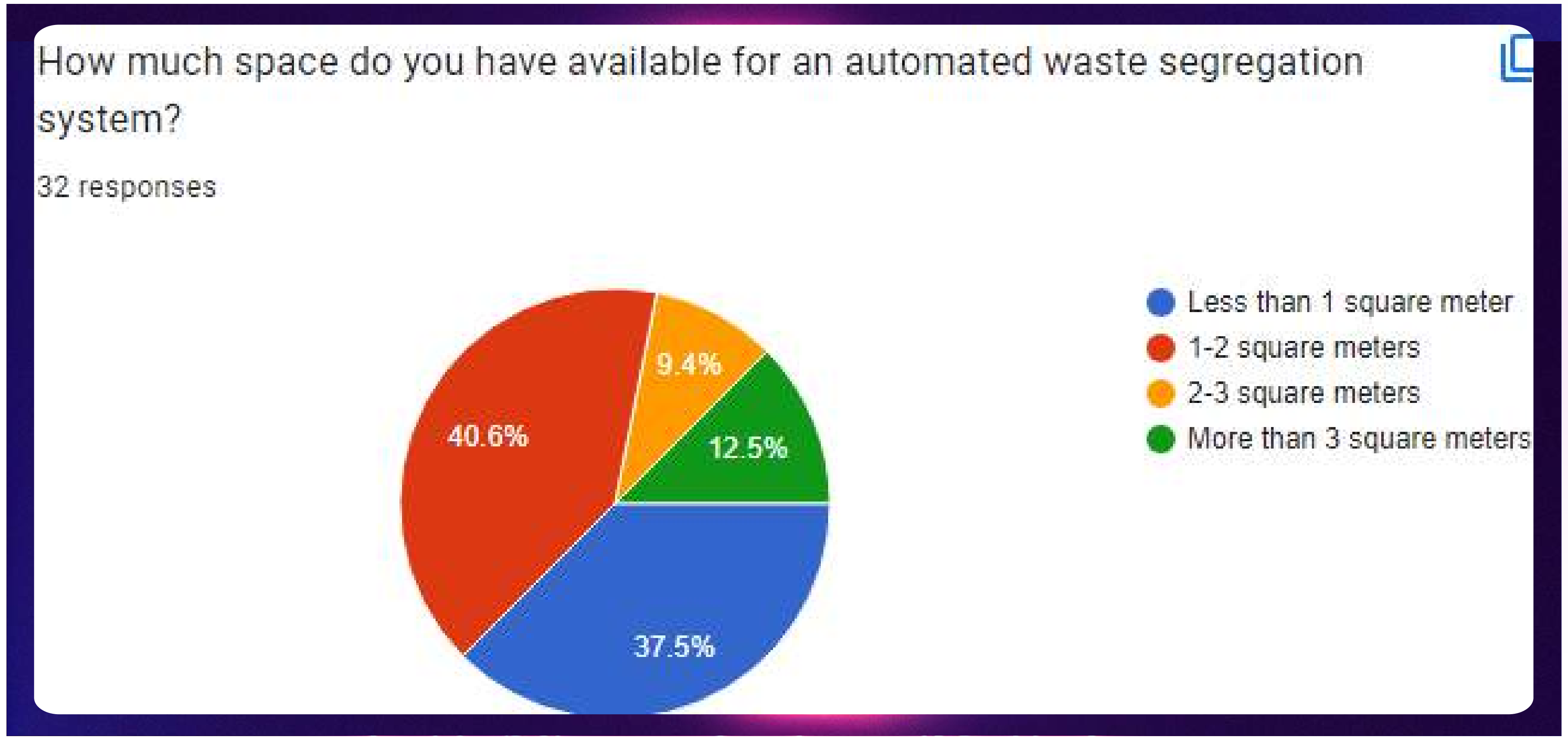

What type of waste do you primarily generate?
32 responses
Household
Industrially
Commercial
15.6%
78.1%

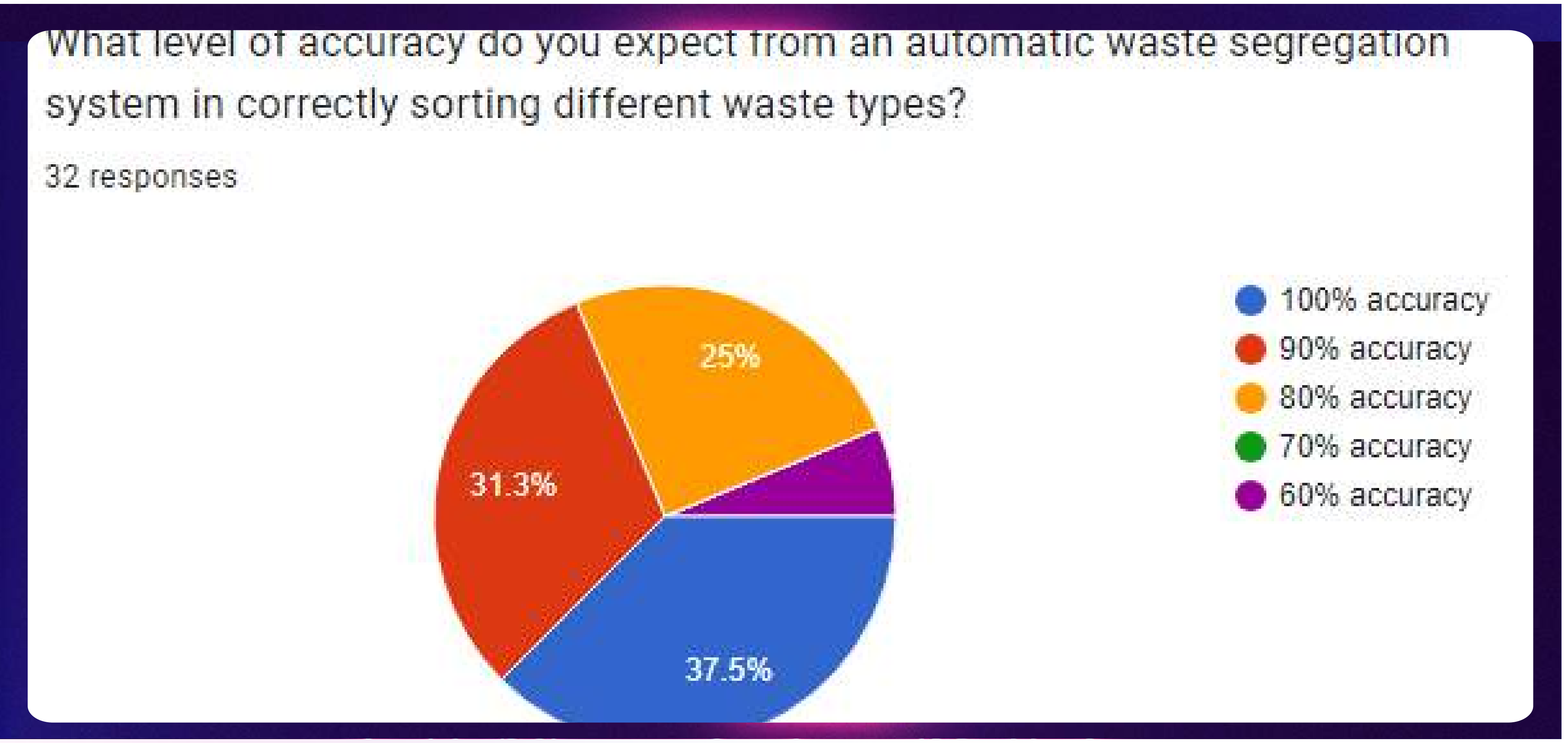

you do segregate waste, what challenges do you face?
Copy cha
responses
Time-consuming
18 (56.3%)
Unpleasant odors
19 (59.4%
Difficulty in identifying recyclable materials
12 (37.5%)
Lack of knowledge about waste segregation
8 (25%)
Health and safety concerns
9 (28.1%)
I do not segregate waste
6 (18.8%)
0
5
10
15
20

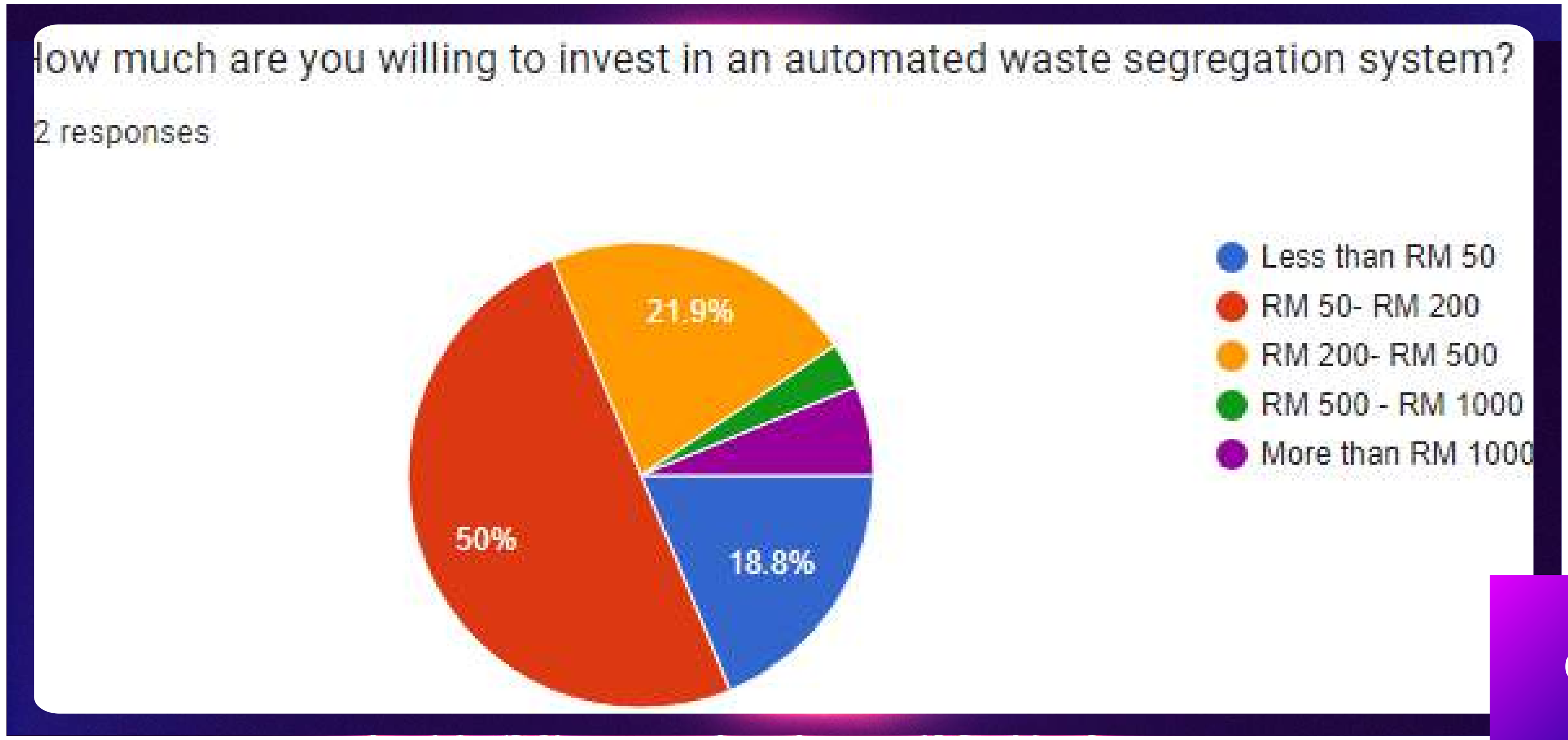

How often do you deal with waste segregation or disposal?
32 responses
Daily
Weekly
Monthly
Rarely
37.5%
15.6%
46.9%


h types of waste are the most challenging for you to segregate?
Copy
ponses
Plastic
15 (46.9%)
Glass
20
Metal
14 (43.8%)
Paper
11 (34.4%)
E-waste
7 (21.9%)
0
5
10
15
20


How important is it for you to have an automated system that can handle multiple types of waste simultaneously?
32 responses
Very important
Important
Moderately important
Slightly important
Not important
56.3%
25%
9.4%
9.4%

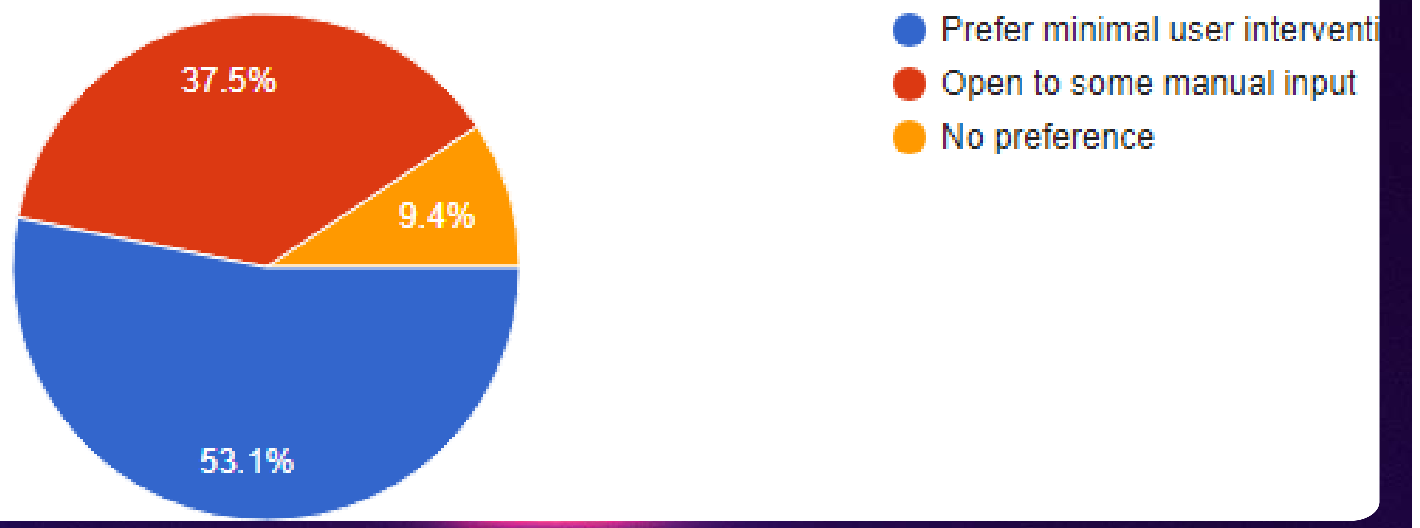

How much space do you have available for an automated waste segregation system?
32 responses
Less than 1 square meter
1-2 square meters
2-3 square meters
More than 3 square meters
37.5%
40.6%
9.4%
12.5%


What level of accuracy do you expect from an automatic waste segregation system in correctly sorting different waste types?
32 responses
100% accuracy
90% accuracy
80% accuracy
70% accuracy
60% accuracy
37.5%
31.3%
25%


low much are you willing to invest in an automated waste segregation system?
2 responses
Less than RM 50
RM 50- RM 200
RM 200- RM 500
RM 500 - RM 1000
More than RM 1000
18.8%
50%
21.9%

ould you prefer a system that requires minimal user intervention, or are you en to systems that require some manual input?

responses



Any recommendations/comments ?

11 responses

no

In moayed we believe

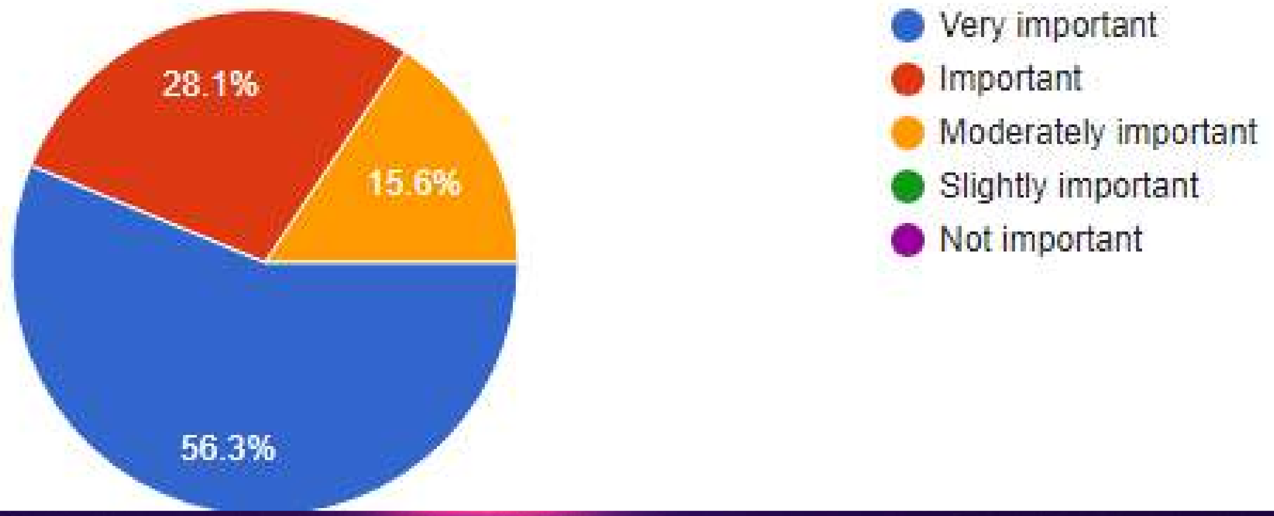
Implement real-time monitoring and data analytics to track system performance and waste segregation efficiency.

IoT and AI

None

How important is it for the system to provide real-time data and feedback on waste segregation performance?

32 responses



make it happen

good and useful survey

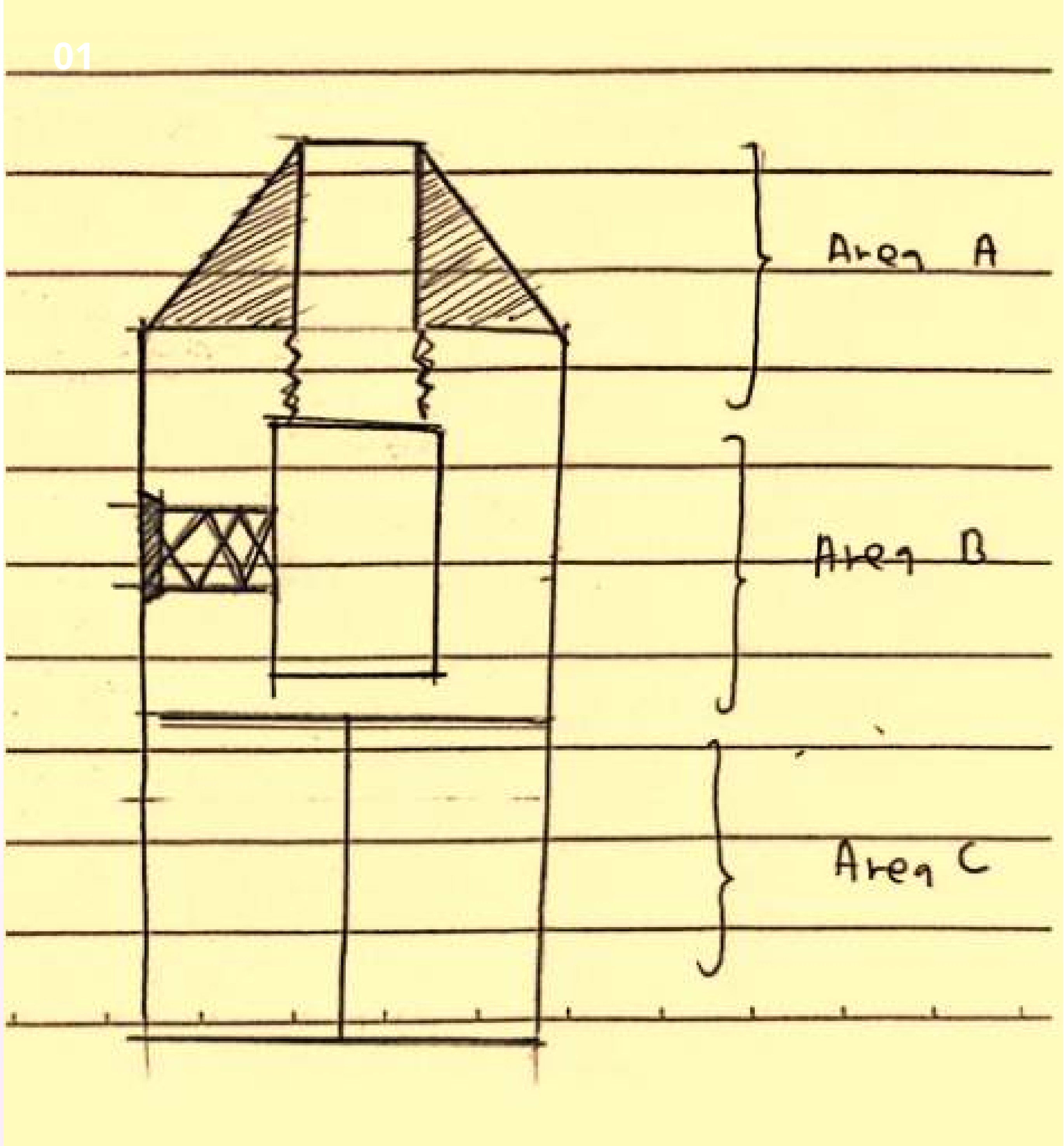
Establish partnerships with local communities, businesses, and waste management org promote awareness, provide resources, and incentivize proper waste segregation practi

Non

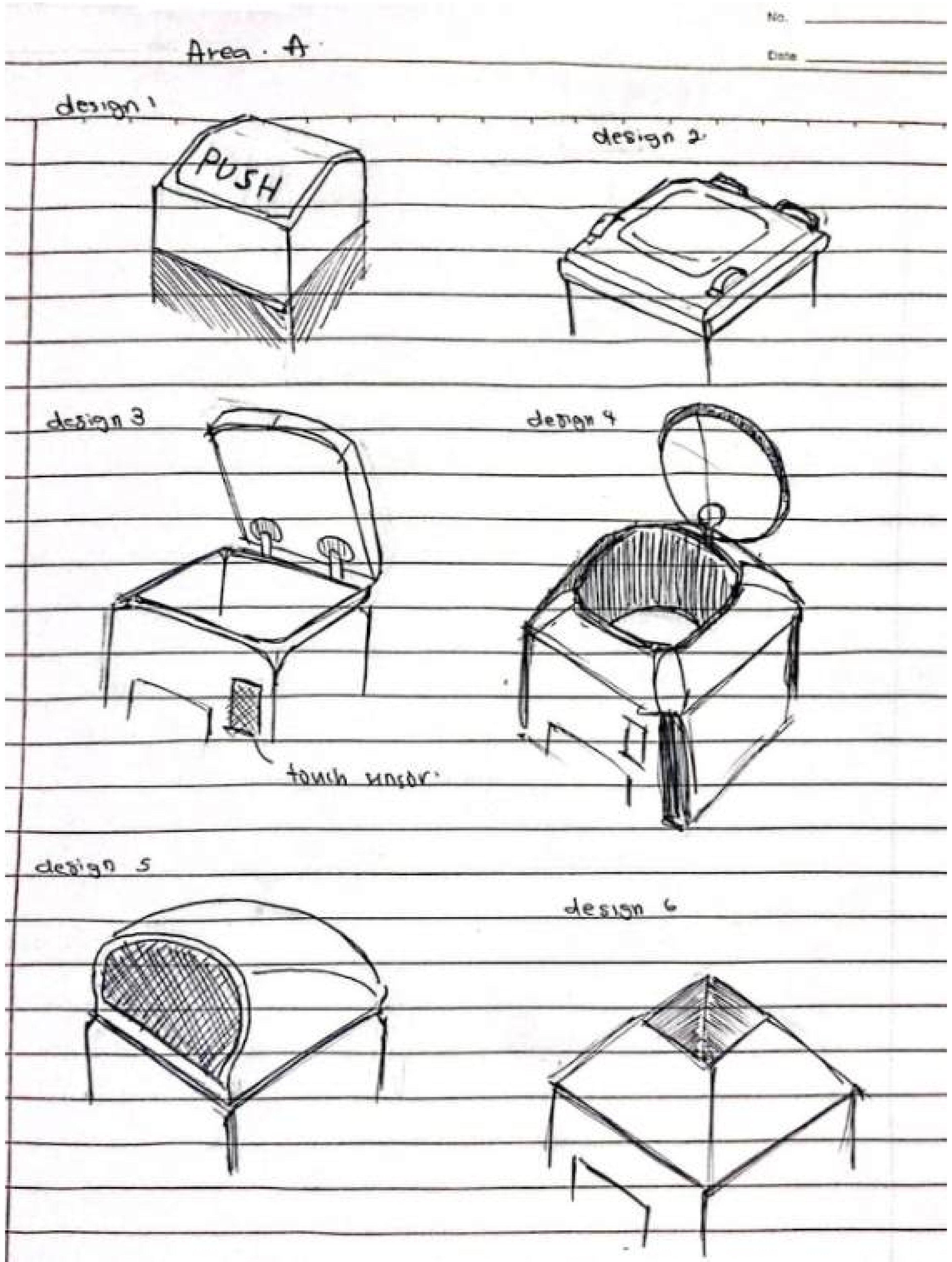
.

Design of The System

Concept Design



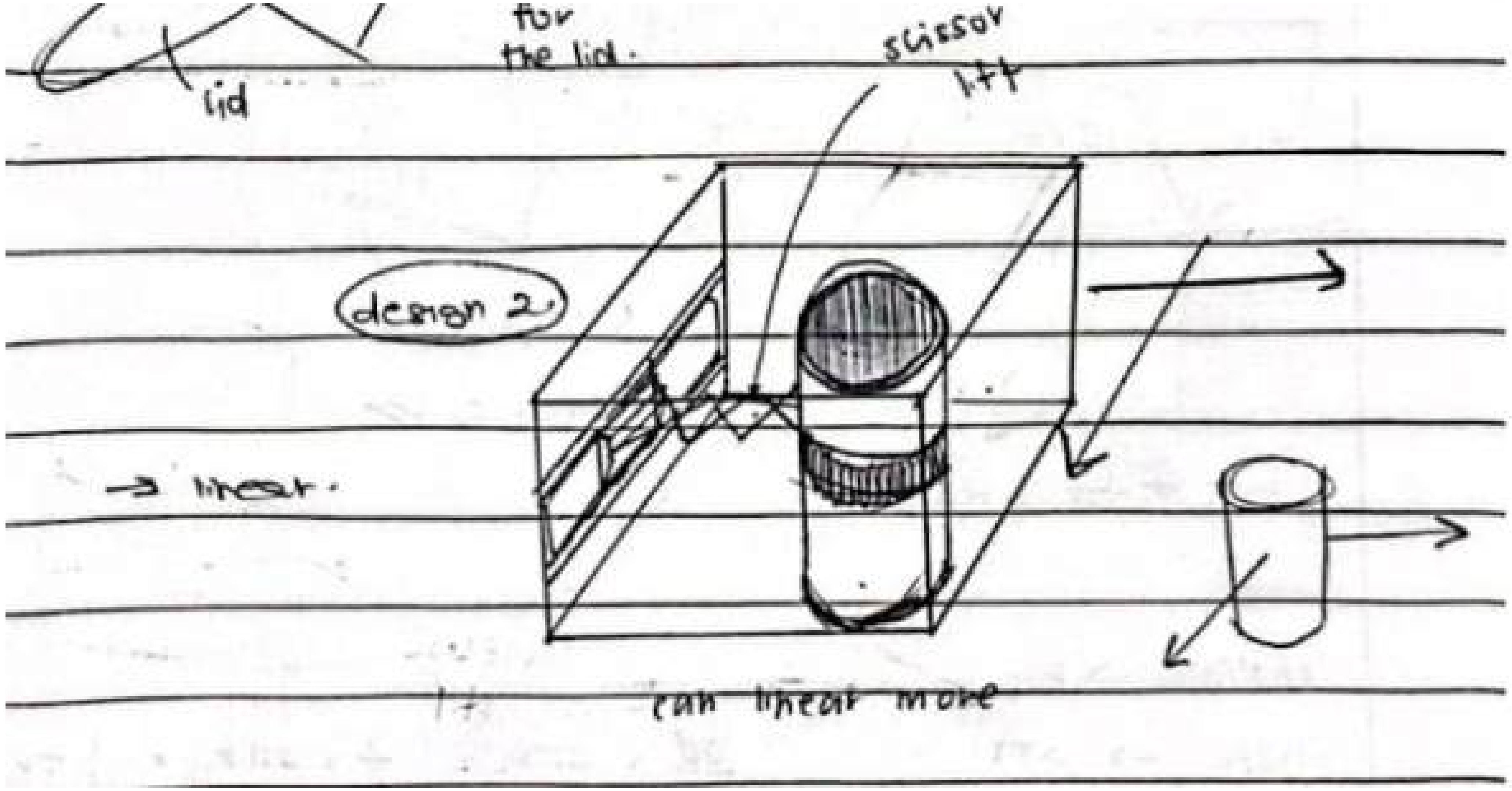
Area A (the head)



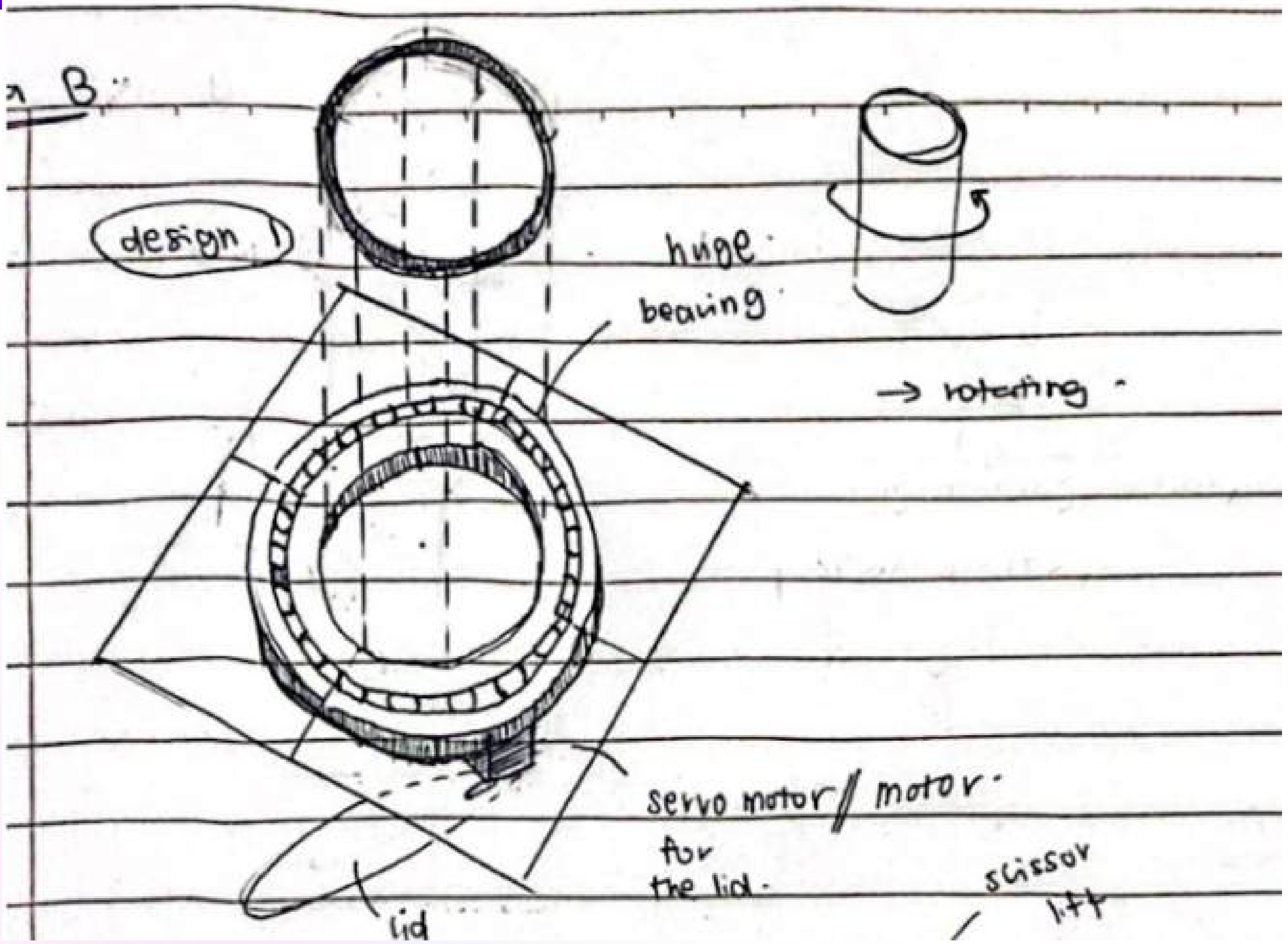
Design of The System

Area B (the body)

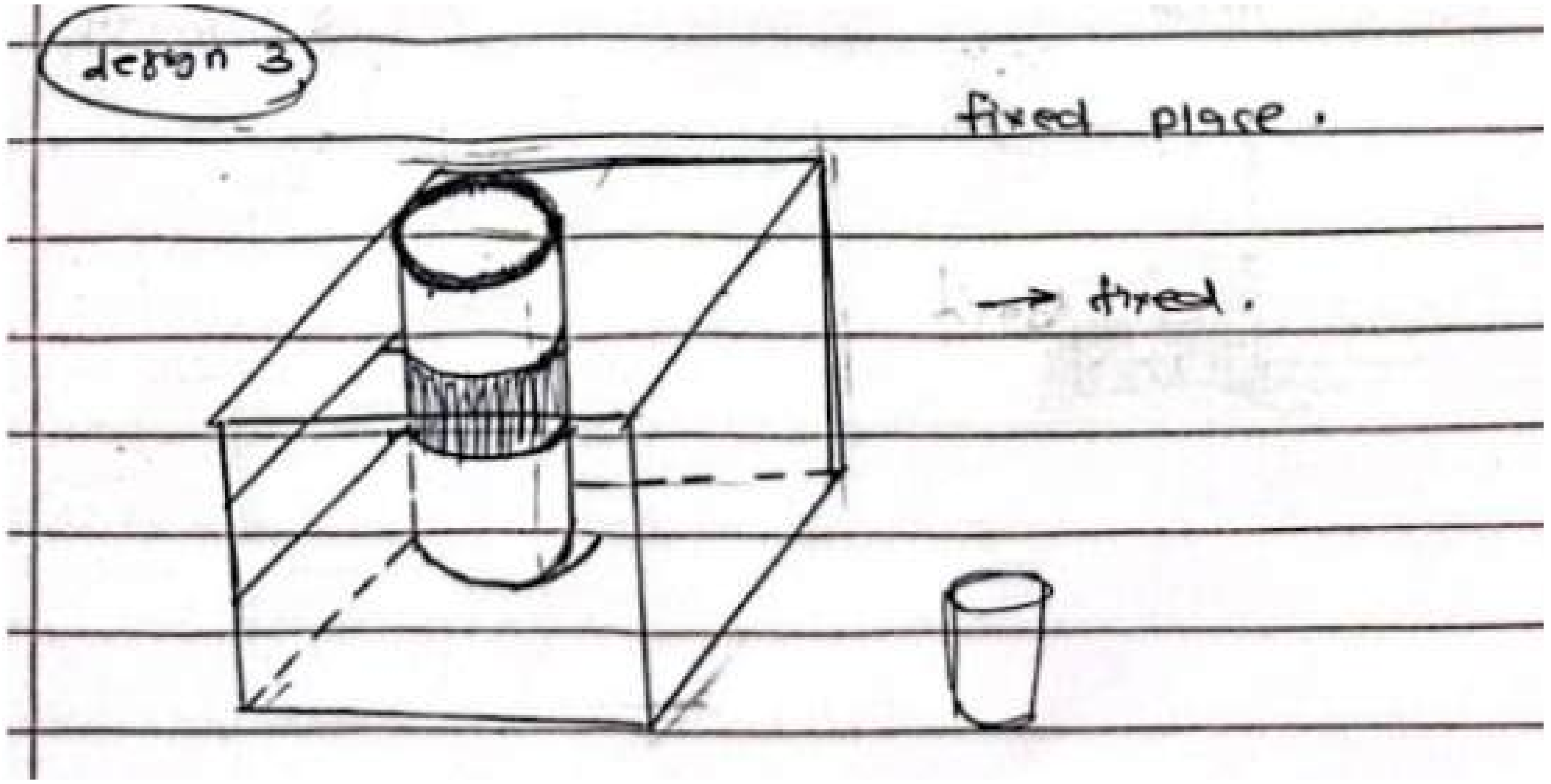
2



1

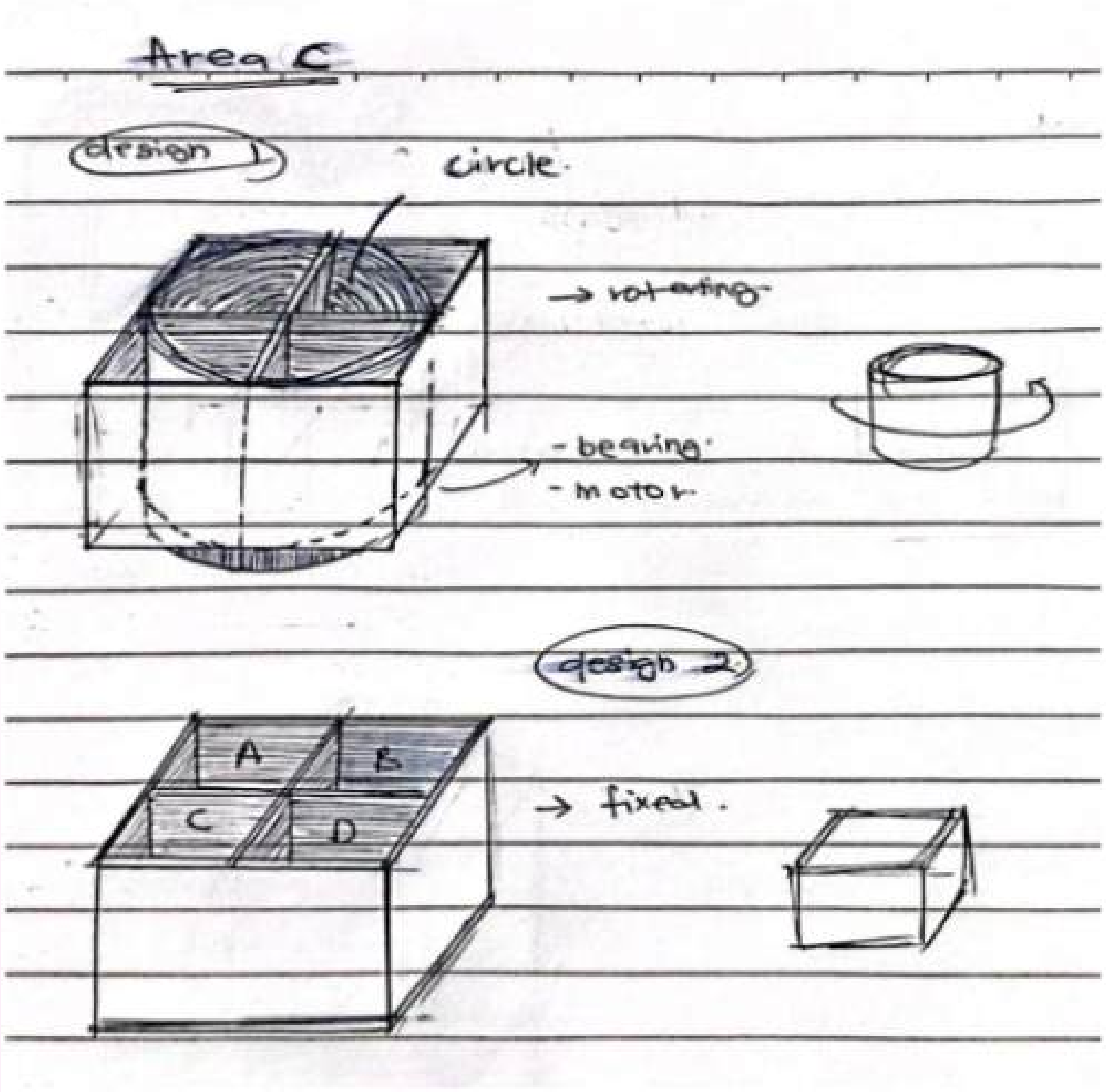


3



Design of The System

Area C (the body)



- **First Combination (Design 1) :**

Area B: Design 2 (linear movement mechanism similar to a 3D printer)

Area C: Design 2 (fixed compartments)

- **Second Combination (Design 2):**

Area B: Design 3 (fixed cylinder)

Area C: Design 1 (rotating compartments with bearing and motor)

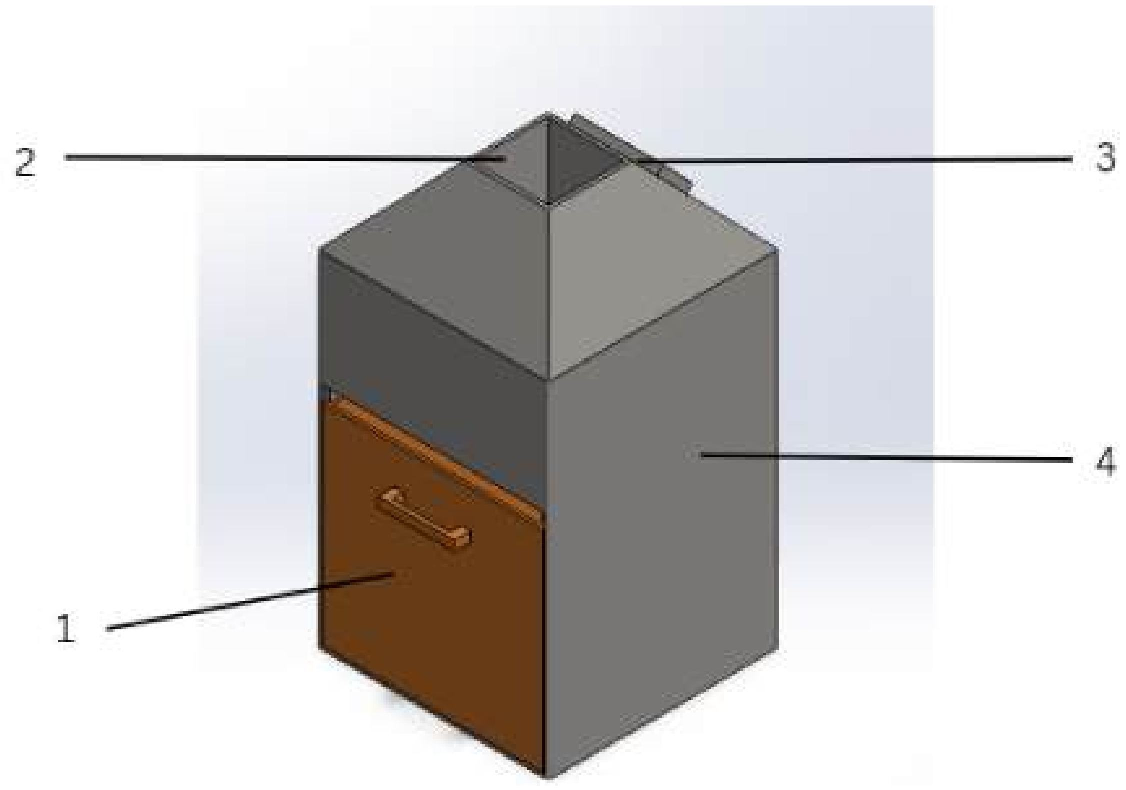
- **Third Combination (Design 3):**

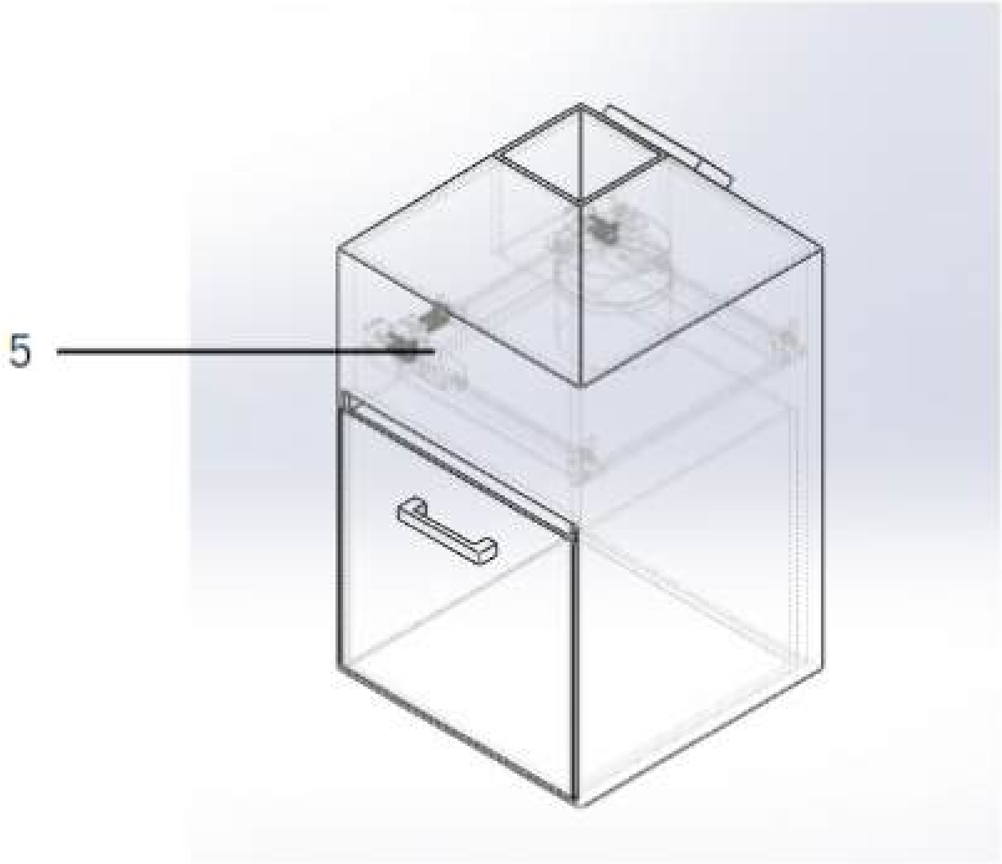
Area B: Design 1 (rotating mechanism with servo motor)

Area C: Design 2 (fixed compartments)

chice is Design 1

FINALIZED PRODUCT

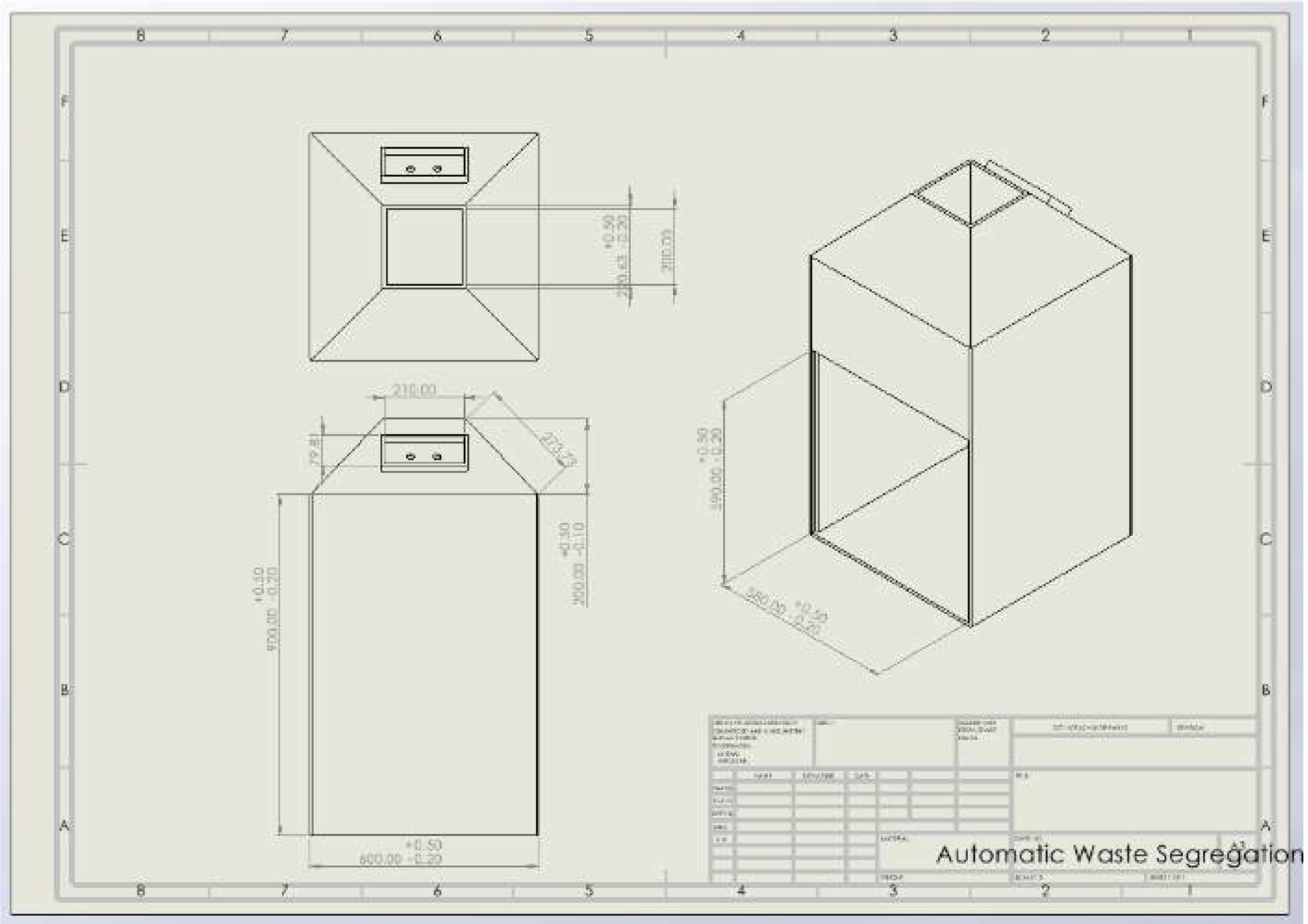




1	**Storage Box**
2	Input Hopper
3	User Interface Section
4	Frame
5	Segregate Mechanism

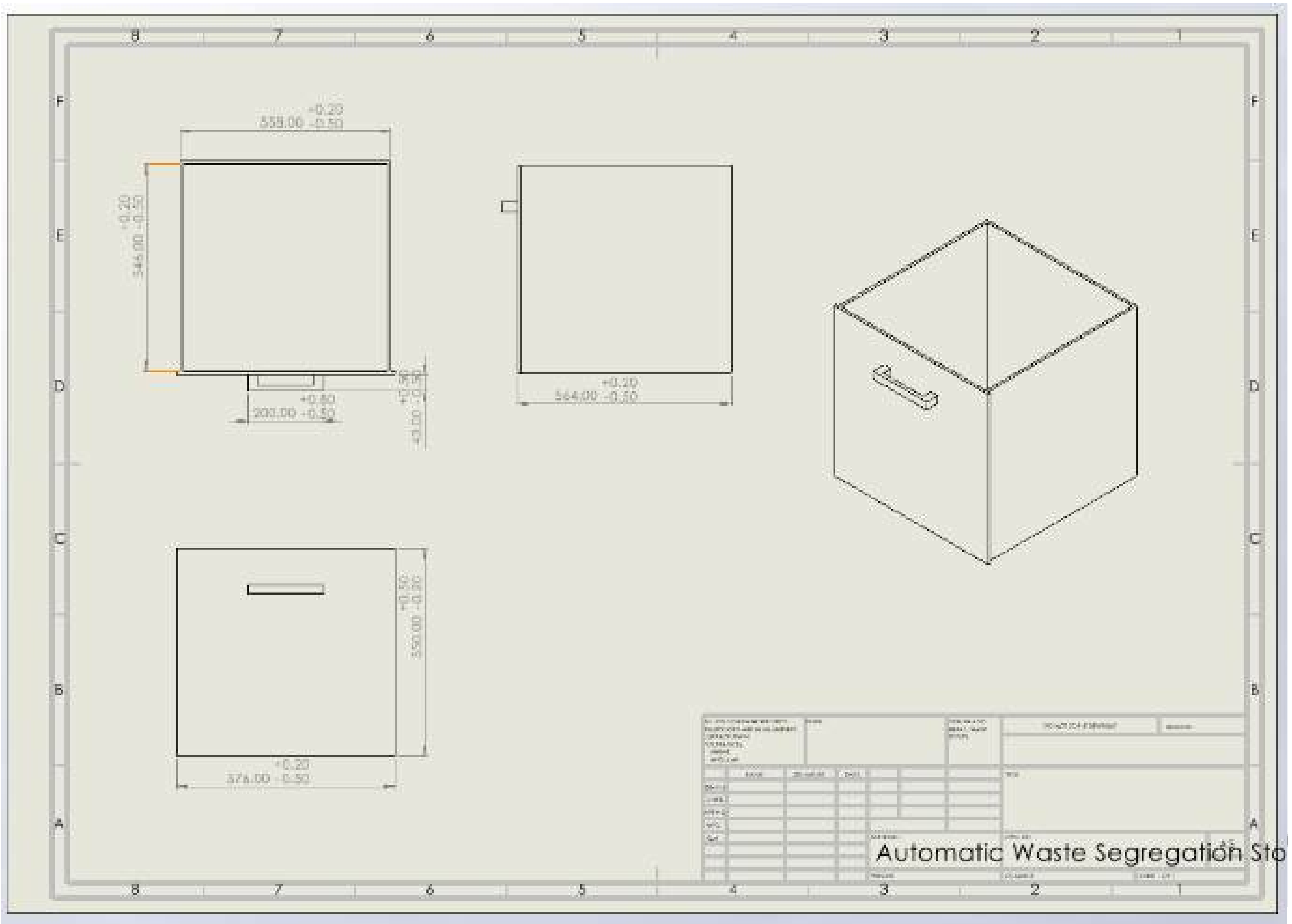
FINALIZED PRODUCT

1. MAIN FRAME



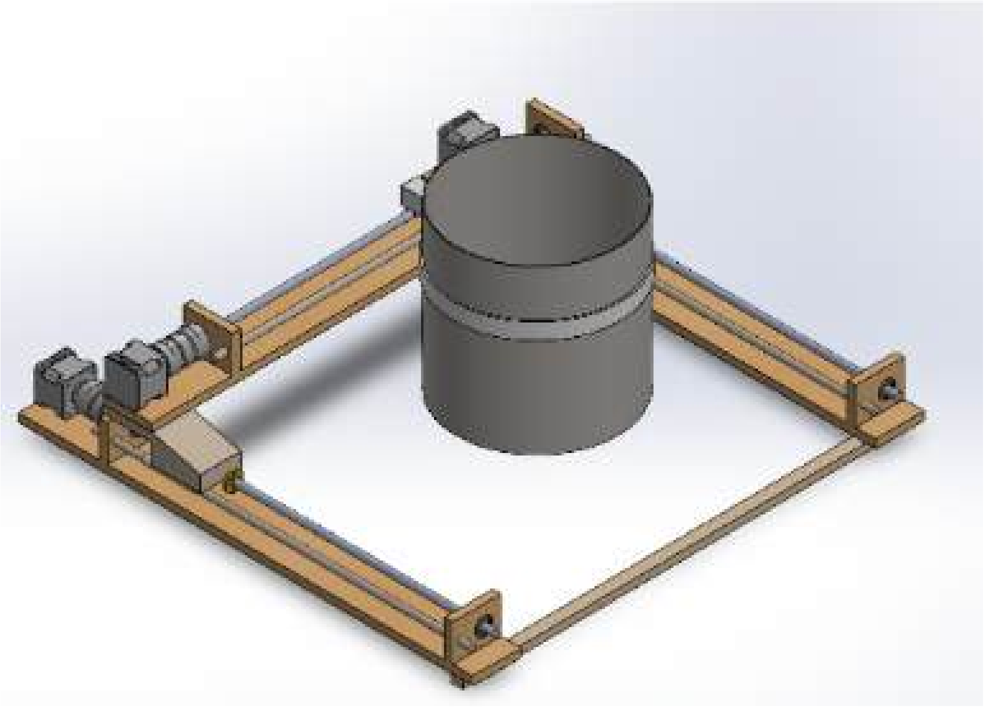
FINALIZED PRODUCT

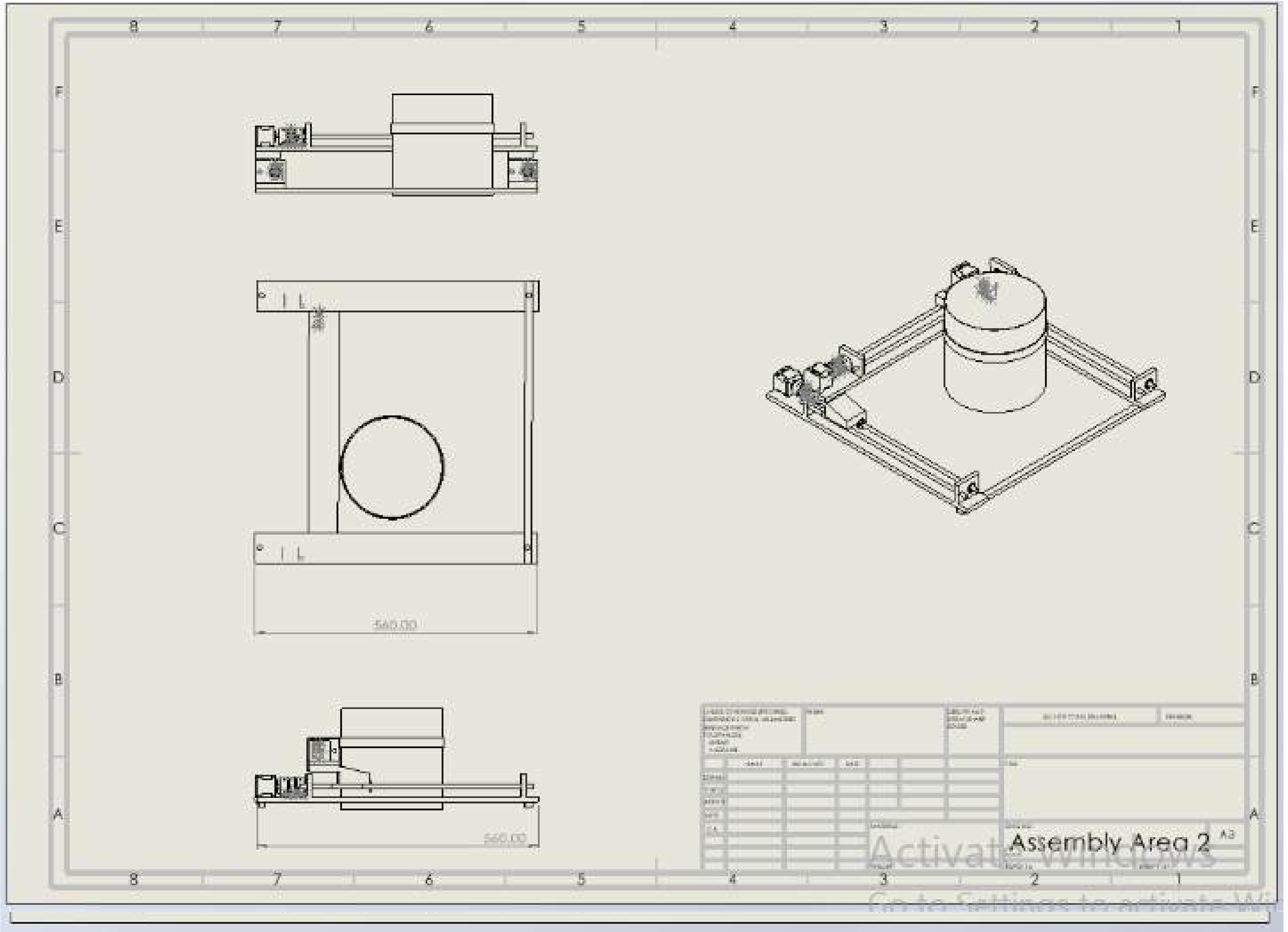
2. STORAGE BOX



FINALIZED PRODUCT

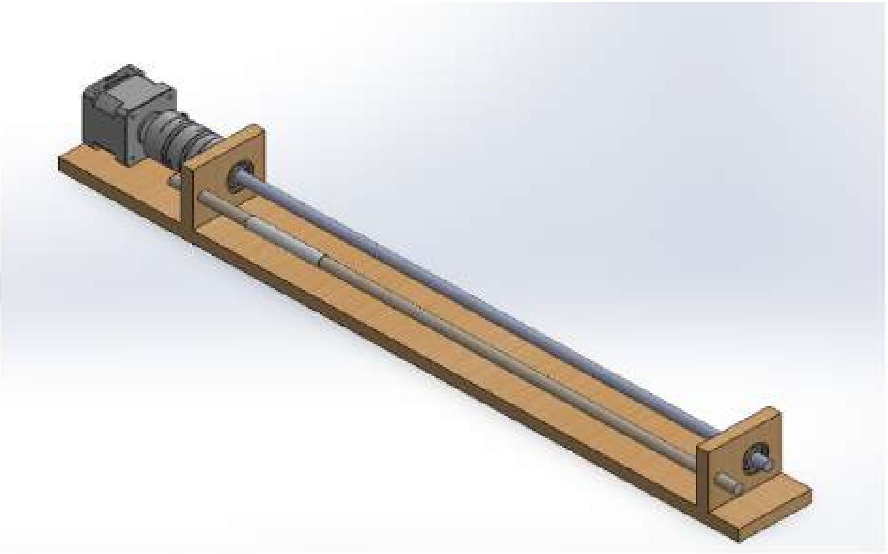
3. SEGREGATE MECHANISM

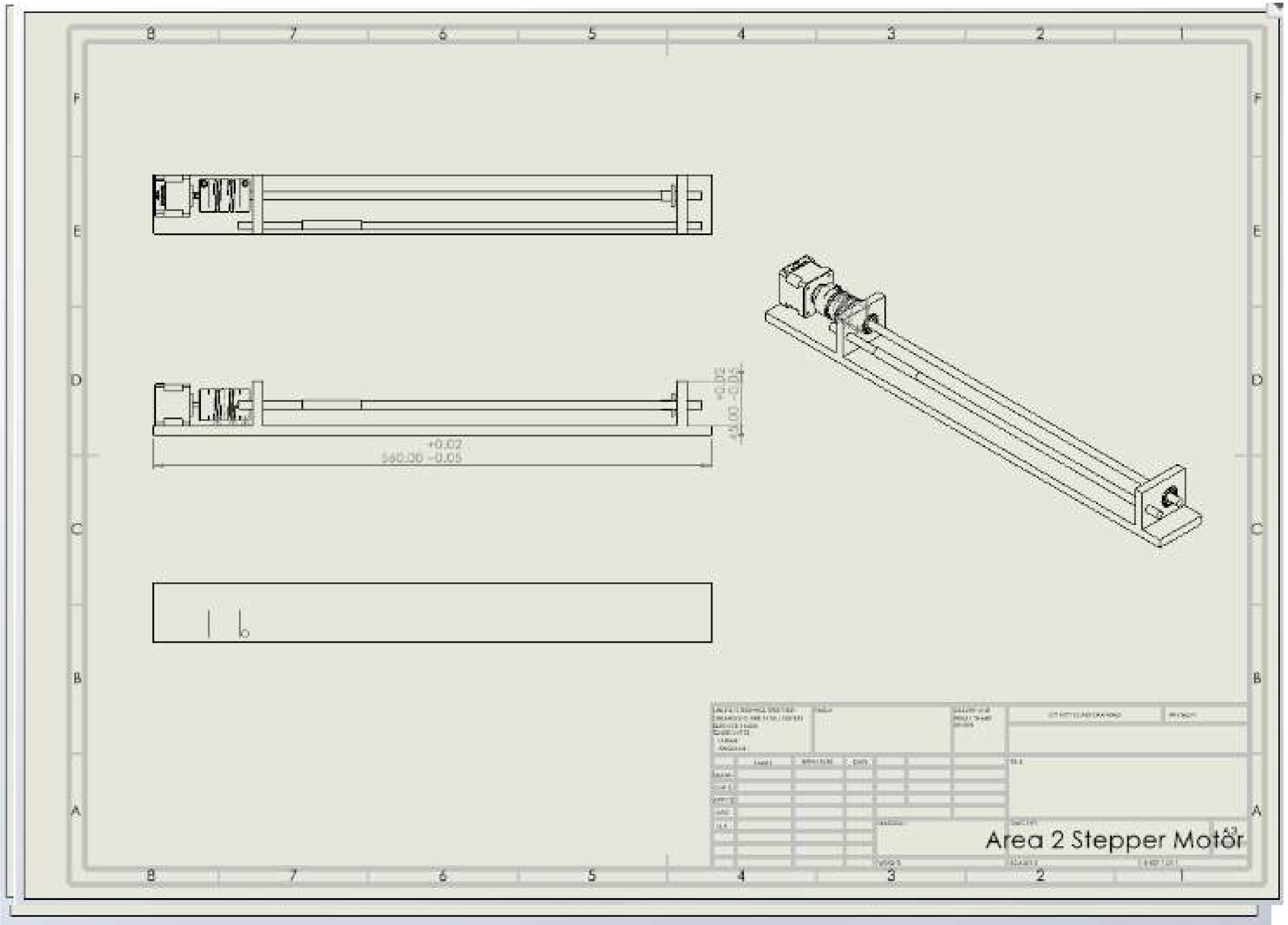




FINALIZED PRODUCT

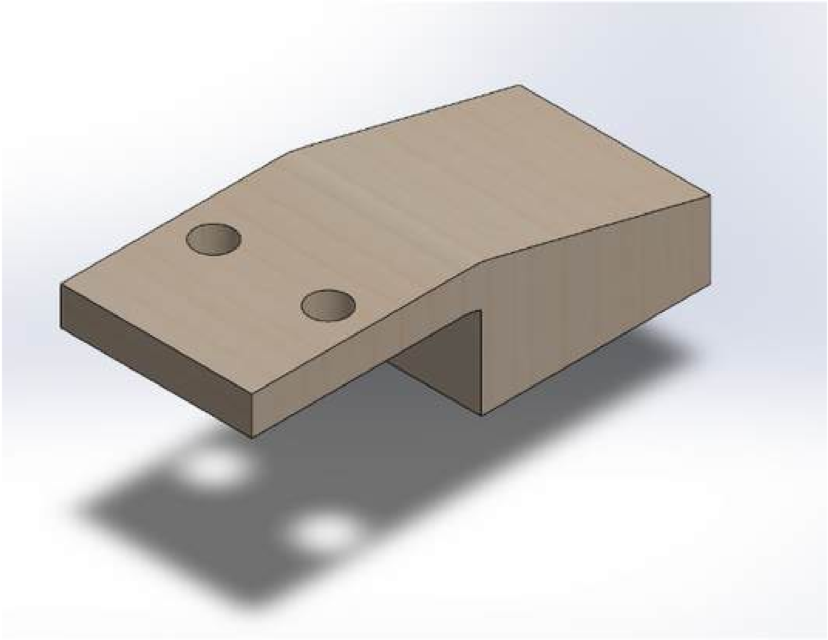
4. STEPPER MOTOR PLATFORM

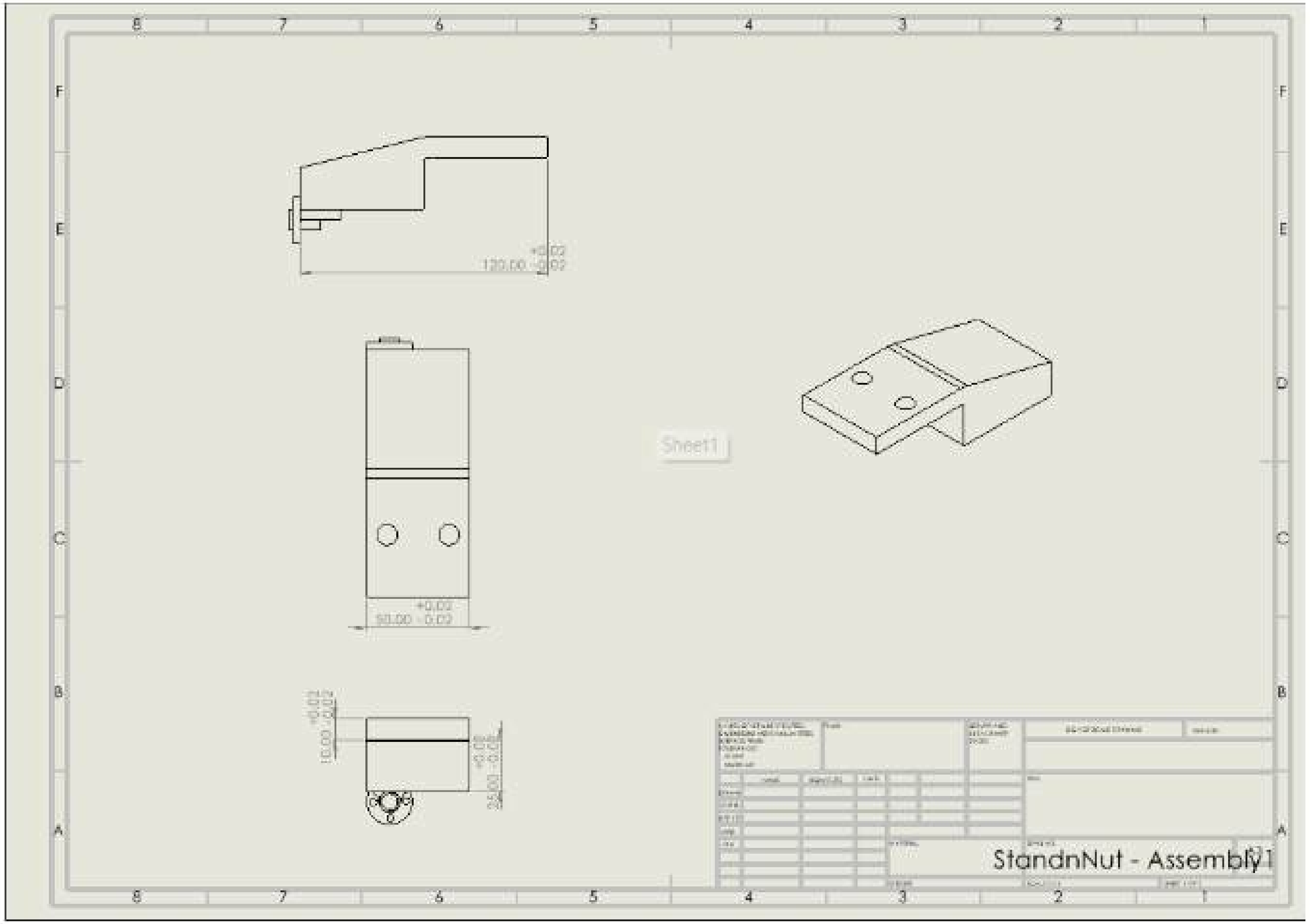




FINALIZED PRODUCT

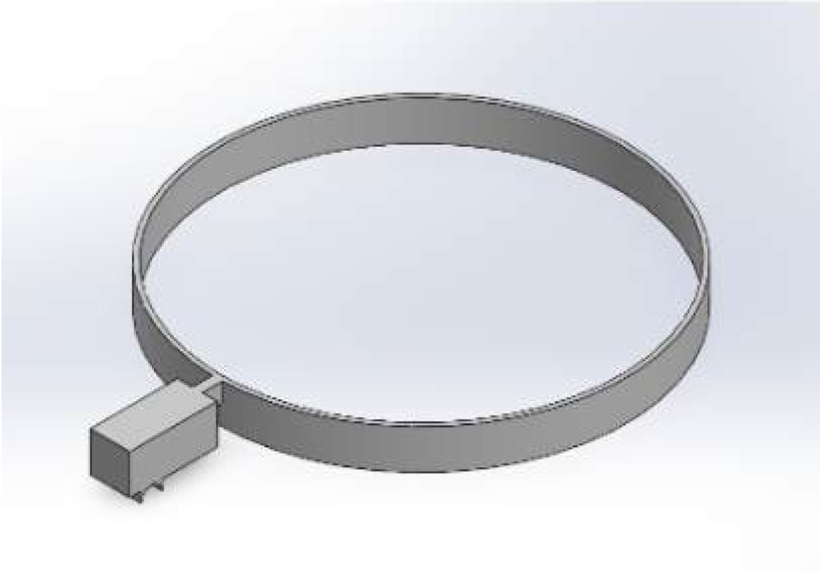
5. STEPPER MOTOR PLATFORM HOLDER

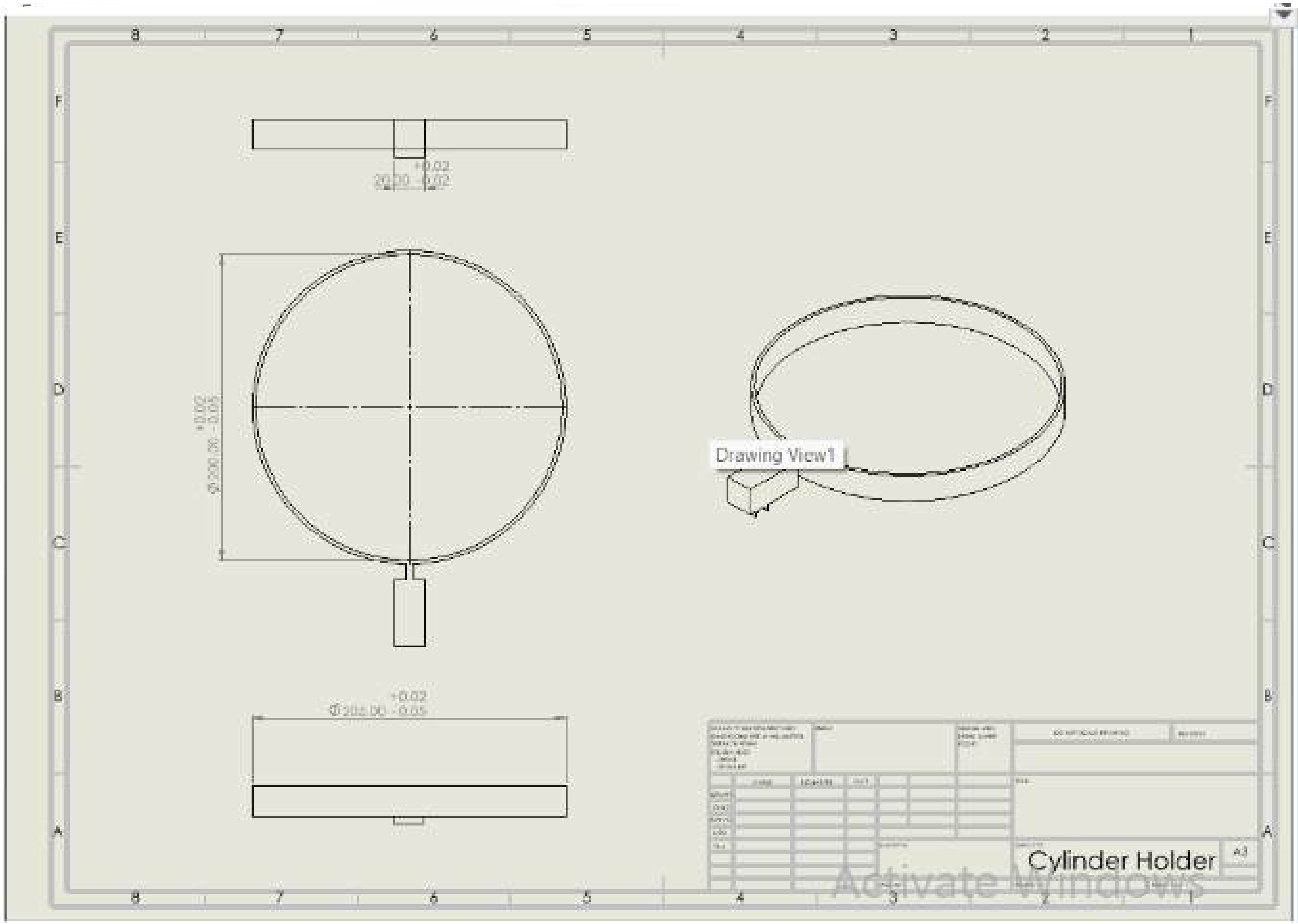




FINALIZED PRODUCT

6. CYLINDER HOLDER (SEGREGATE MECHANISM)





LIST ITEM

Microcontroller	• **Model** : Arduino Mega 2560 • Processor : ATmega2560 • Digital I/O Pins : 54 (15 PWM outputs) • Analog Input Pins : 16 • Operating Voltage : 5V • Flash Memory : 256 KB • SRAM : 8 KB
Motor driver	• **Model :** DRV 8825 Stepper Motor Driver • Voltage : 8.2V to 45V • Current : 1.5A per phase (2.2A peak) • Microstepping : Up to 1/32 step • Interface : Step and direction control
Bipolar Stepper Motor	• **Model :** NEMA 17 • Step Angle: 1.8° per step • Holding Torque: 45 N·cm (63.74 oz·in) • Current: 1.5A per phase • Voltage: 12V
Servo Motor	• **Model :** MG996R • Operating Voltage: 4.8V - 7.2V • Stall Torque: 9.4 kg·cm (4.8V), 11 kg·cm (6V) • Speed: 0.19 sec/60° (4.8V), 0.14 sec/60° (6V) • Rotation: 180°

Ultrasonic sensor	• **Model :** HC-SR04 • Operating Voltage: 5V • Measuring Range: 2cm - 400cm • Resolution: 0.3cm • Angle: 15 degrees

Wifi module	• **Model :** ESP8266 • Operating Voltage: 3.3V • Wi-Fi Standard: 802.11 b/g/n • Frequency Range: 2.4GHz - 2.5GHz • Flash Memory: 4 MB
Led	• **Model :** 5mm LED • Operating Voltage: 2V - 3V • Current: 20mA • Colours: Red, Green, Blue (RGB option available)
LCD	• **Model :** 16x2 LCD (HD44780) • Operating Voltage: 5V • Interface: Parallel • Backlight: LED backlight • Character Size: 5x8 dots
Wire	• **Type:** Black Red Wire • Length: 20m • Gauge: 22 AWG
Frame	• **Material :** Stainless Steel or Durable Plastic • 600mm x 600mm x 1000mm (cuboid shape) • Thickness : 20mm x 20mm • The whole main body
Breadboard	• Size: 830 tie-points • Dimensions: 16.5 x 5.4 x 0.85 cm • Rows: 2 power rails, 63 rows • Columns: 10

LIST ITEM

Camera	• **Model :** Raspberry Pi Camera Module V2 • Resolution: 8 Megapixels • Video: 1080p at 30fps • Interface: CSI connector
Power Adapter	• Input: 100-240V AC, 50/60Hz • Output: 12V DC, 2A • Connector: Barrel Jack
Lids	• **Material:** Stainless Steel or Durable Plastic • Shape : Square • Automatic with Touch Sensor
Touch Sensor	• **Model:** TTP223 Capacitive Touch

Battery Backup	• **Model:** Li-Ion Rechargeable Battery Pack • Voltage: 12V • Capacity: 2000mAh • Output Current: 1A
Microcontroller shield	• **Model :** Arduino Mega Shield for stepper motor drivers (CNC Shield V3) • Compatible with Arduino Mega 2560 • Designed for easy connection of multiple stepper motor drivers, such as the DRV8825 • Voltage Range: Typically 12-36V DC • Jumpers to set microstepping resolution (up to 1/32 step with DRV8825) • Supports up to 4 stepper motors (X, Y, Z, and an additional axis)
Bearing	• **Model :** 608ZZ • Inner diameter : 8mm • Outer Diameter (OD): 22mm • Width: 7mm • Material: Chrome Steel • Shield: Metal Shields (ZZ) or Rubber Seals (RS) • Load Rating: Dynamic: 3.6 kN, Static: 1.55 kN • Max Speed: 34,000 RPM (for ZZ), lower for RS due to sealing friction • Usage: Suitable for high-speed, low-noise applications; commonly used in linear motion systems.

LIST ITEM

21	Flexible Shaft Coupling	• **Model :** Aluminum Flexible Shaft Coupling • Outer Diameter (OD): 19mm • Length (L): 25mm • Material: Aluminium Alloy • Flexibility: Can accommodate slight angular, parallel, and axial misalignments • Torque Rating: Up to 2 Nm • Usage: Connects motor shafts to lead screws or other shafts, reduces misalignment stress.
22	Round Wooden Stick	• **Material :** Hardwood • Diameter: 8mm (matching the coupling and bearing specifications) • Length: 500mm • Finish: Sanded smooth, optionally treated with a clear finish or varnish for durability • Usage: Can be used as a non-metallic, lightweight alternative for structural or mechanical components.

COSTING

szsmtsw.my >

2pcs Black GT2 Timing Pulley 20 Tooth 1...
20T W6 B8 With T x2
15 Days Free Returns* Pre–Order
RM6.36

GT2 Closed Loop Timing Belt Rubber 2G...
1220mm x1
15 Days Free Returns*
RM11.72

Order Total: RM24.44

Preferred+ AT ALUMINIUM >

T8 Lead Screw Thread diameter 8mm Pit...
8mm,500mm x3
15 Days Free Returns*
~~RM27.90~~ RM25.11

Aluminium Profile 2020 10CM / 20CM / ...
100cm x3
15 Days Free Returns*
~~RM25.00~~ RM23.00

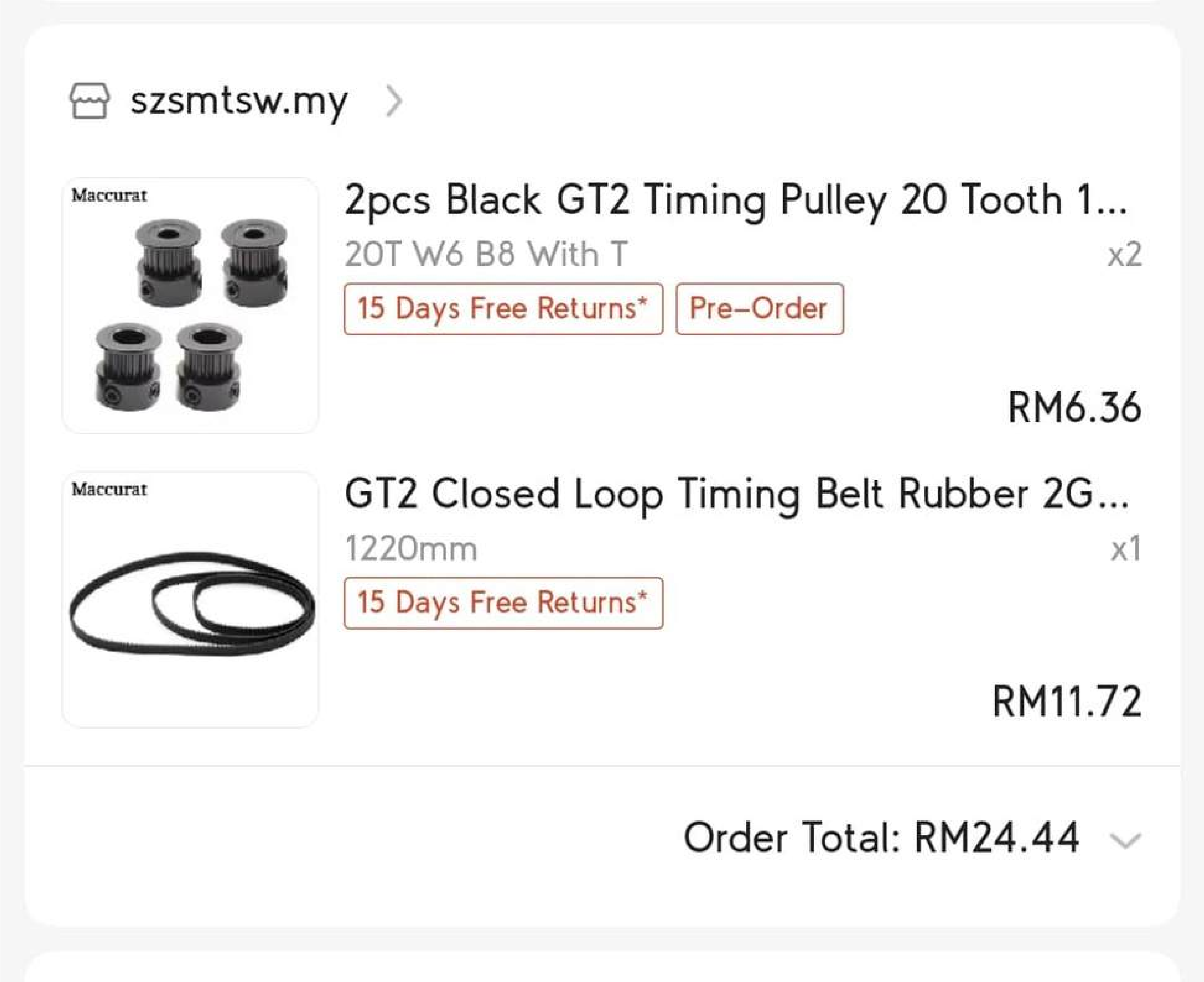
Aluminium Shaft Coupling Flexible Coupl...
5mm – 8mm x6
15 Days Free Returns*
~~RM5.90~~ RM5.61

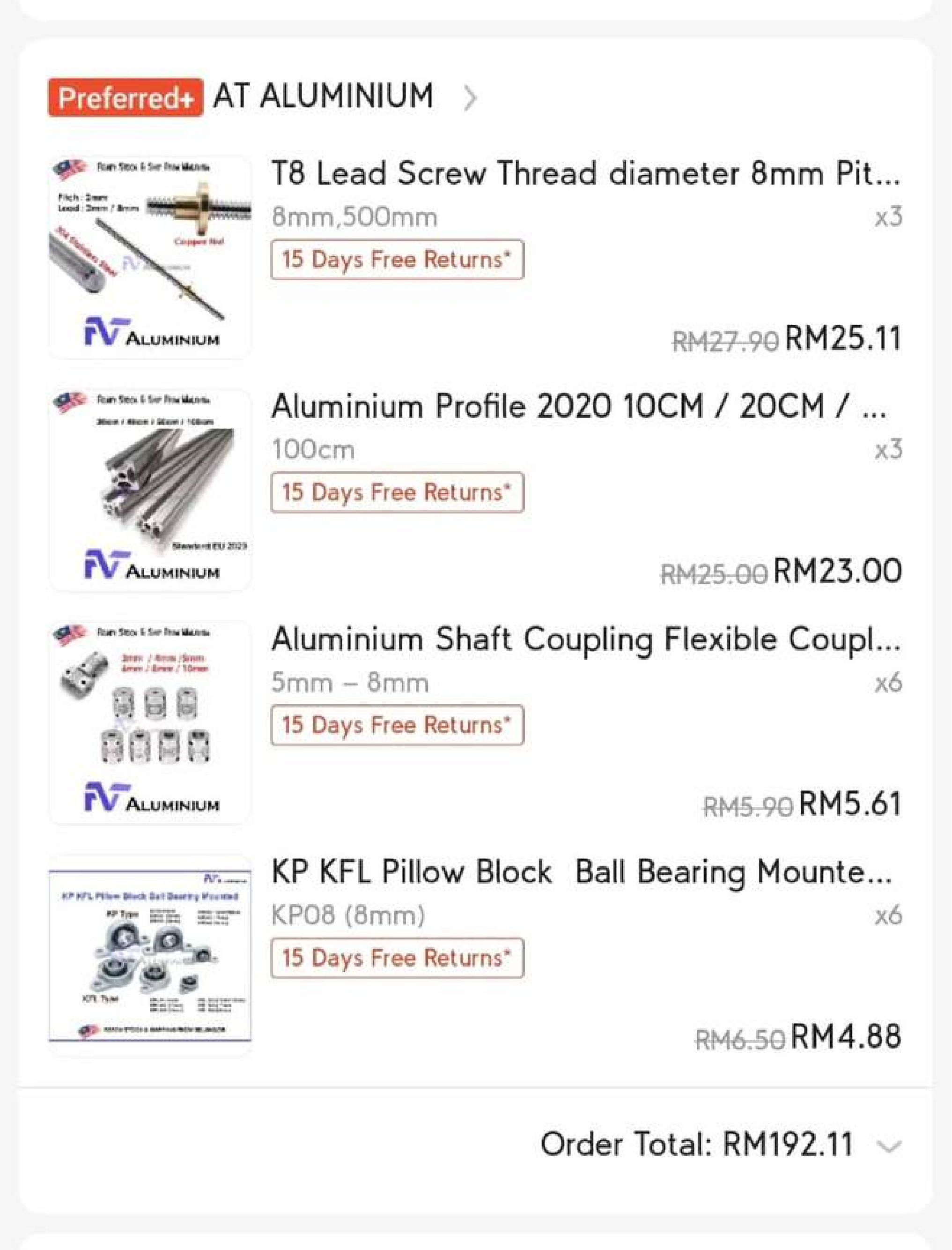
KP KFL Pillow Block Ball Bearing Mounte...
KP08 (8mm) x6
15 Days Free Returns*
~~RM6.50~~ RM4.88

Order Total: RM192.11

TOTAL : RM 541.99

COSTING





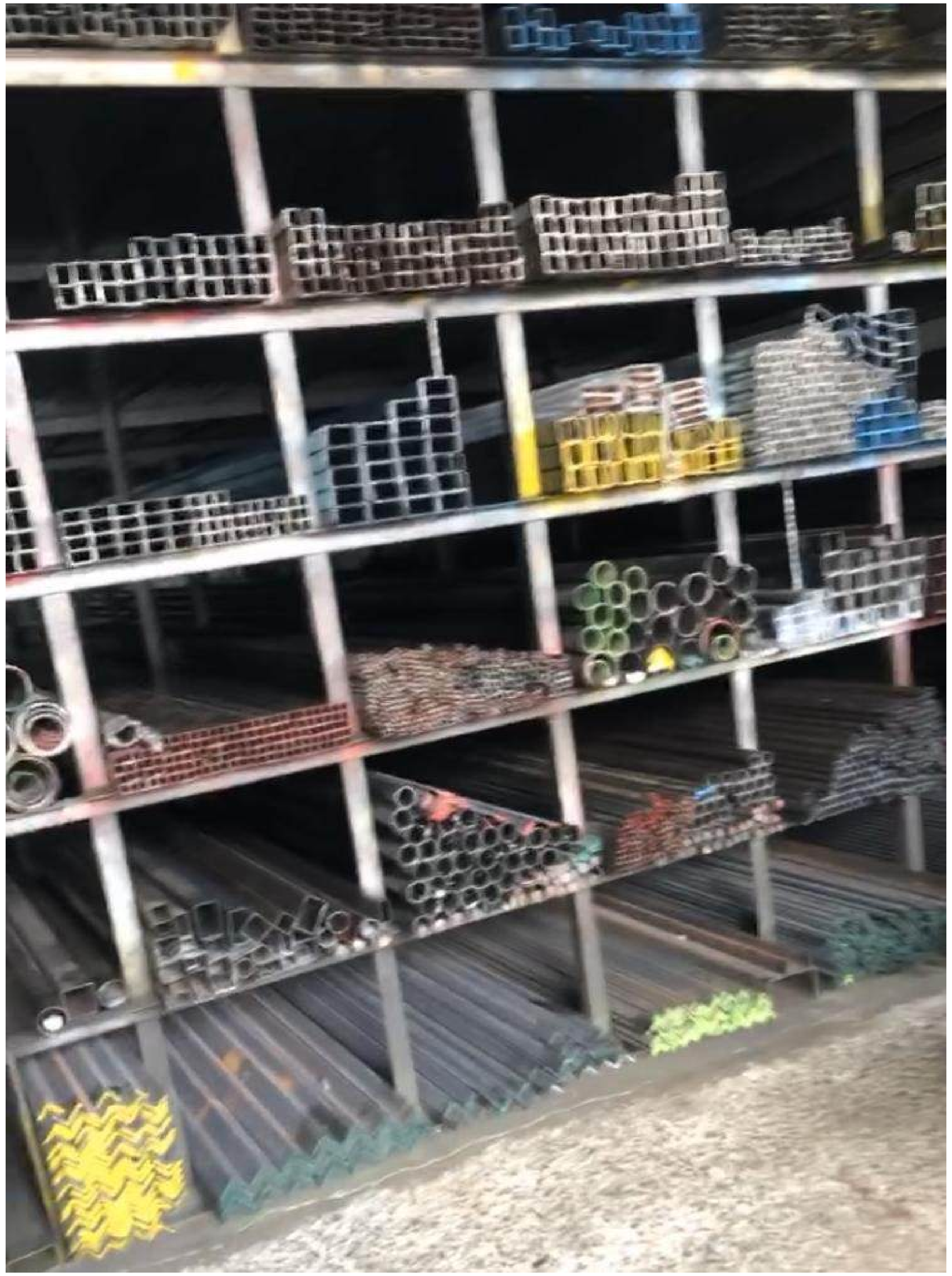
TOTAL : RM 309

progression

- Frame
- 3D printing
- Segregation Mechanism
- calculation

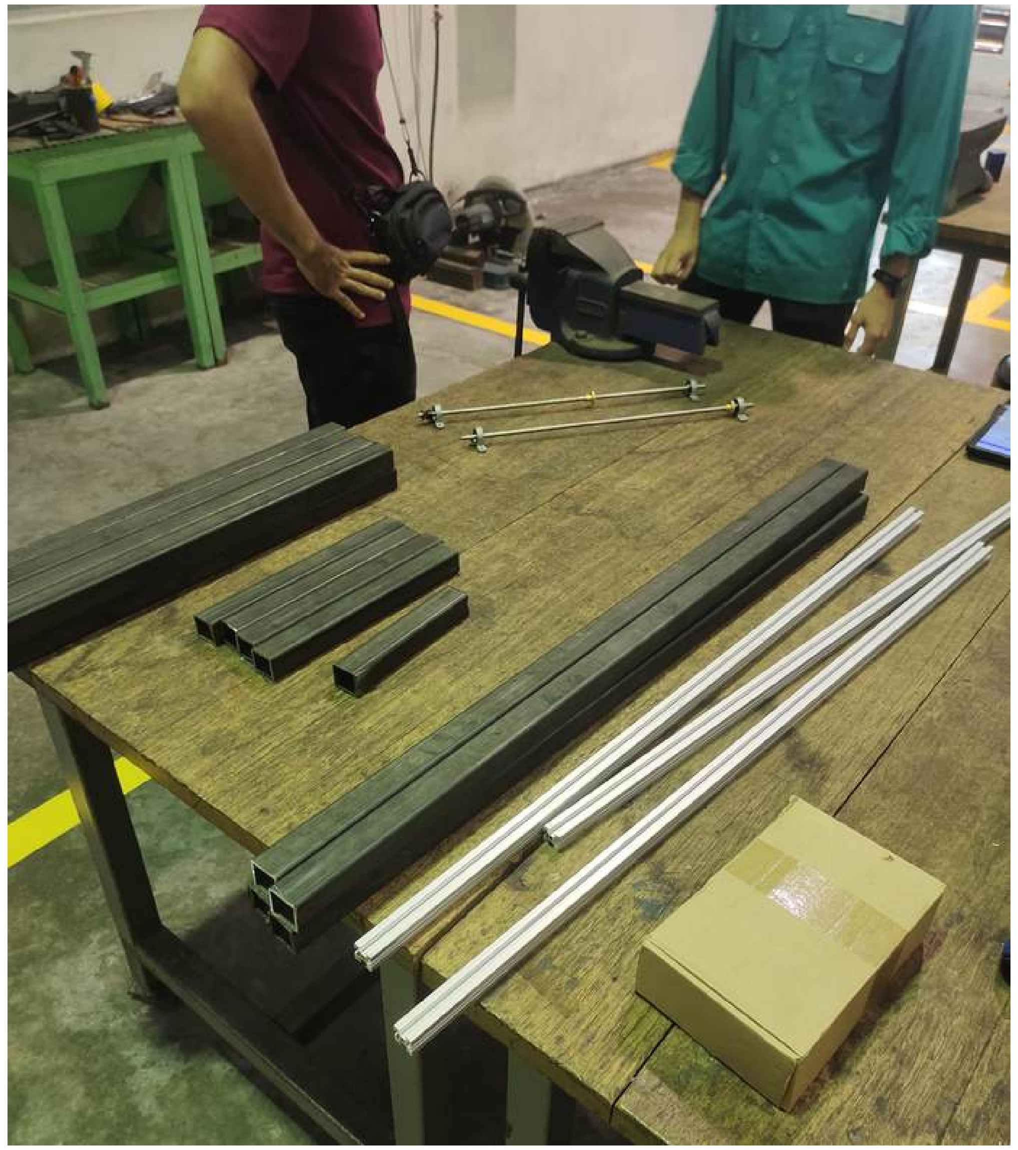
1. Frame

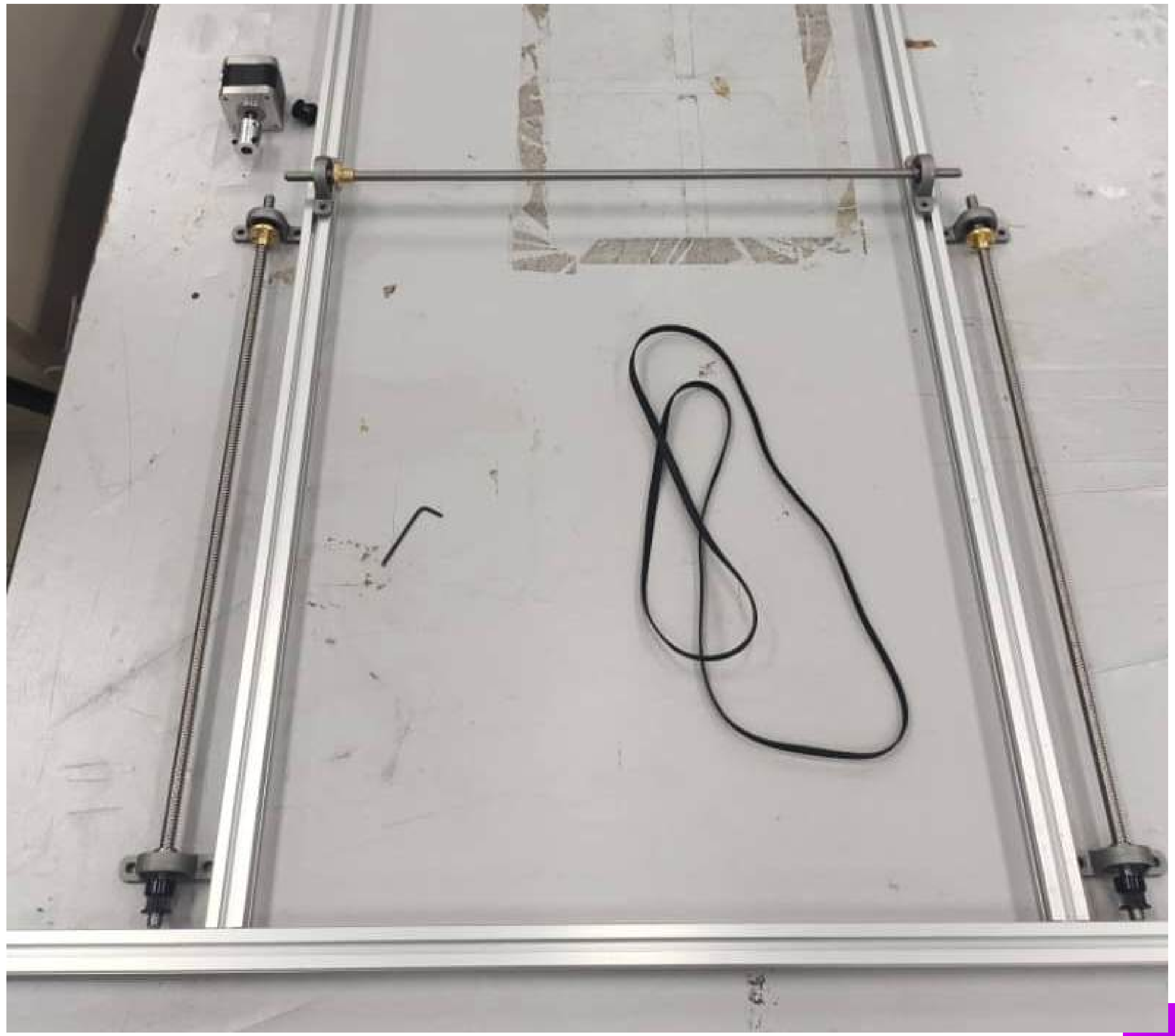
- Material: Stainless Steel
- 600mm x 600mm x 1000mm (cuboid shape)
- Thickness : 30mm x 30mm
- The whole main body



frame processing

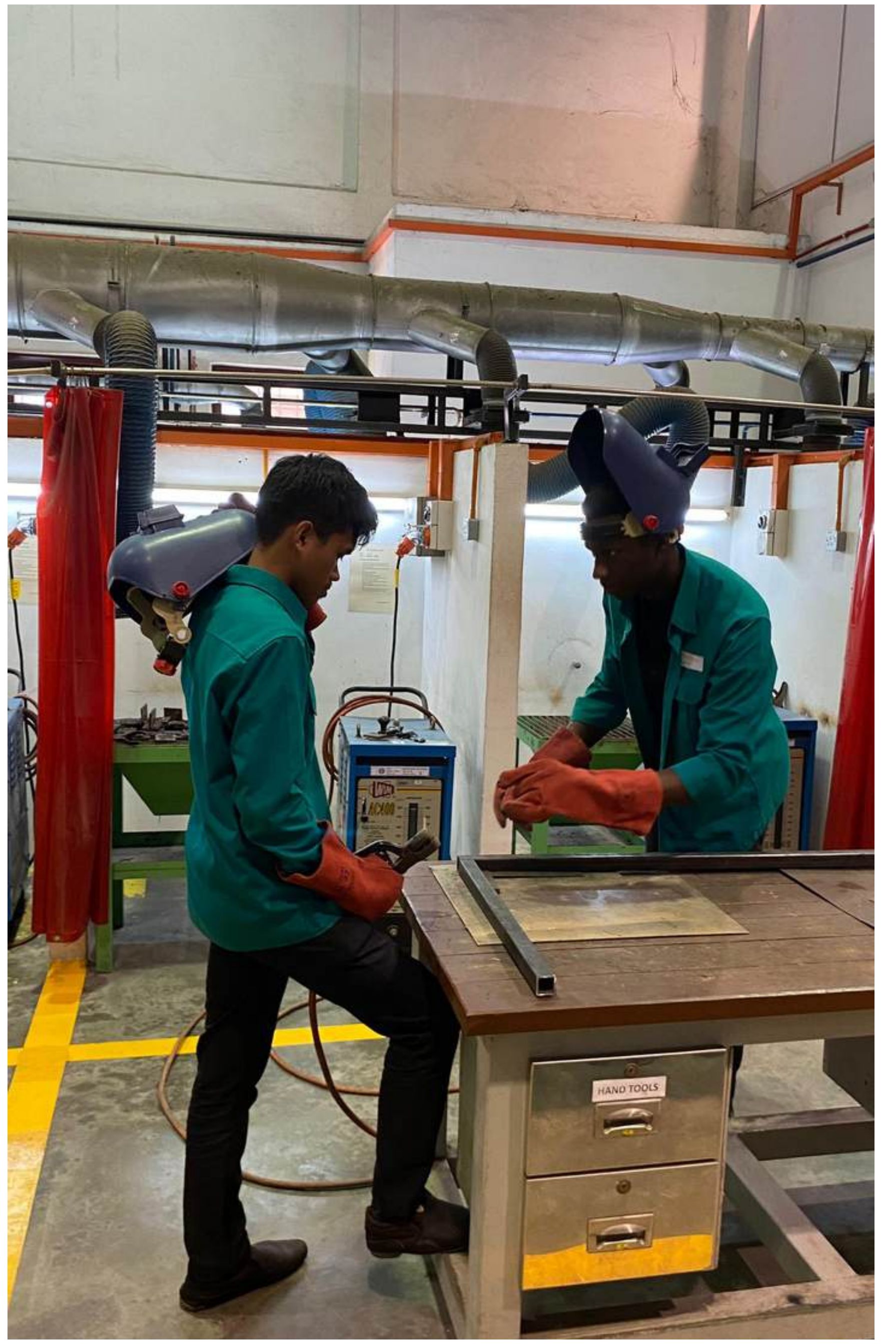
1. Measure Dimension





frame processing

2. Cutting & Welding Metal Components

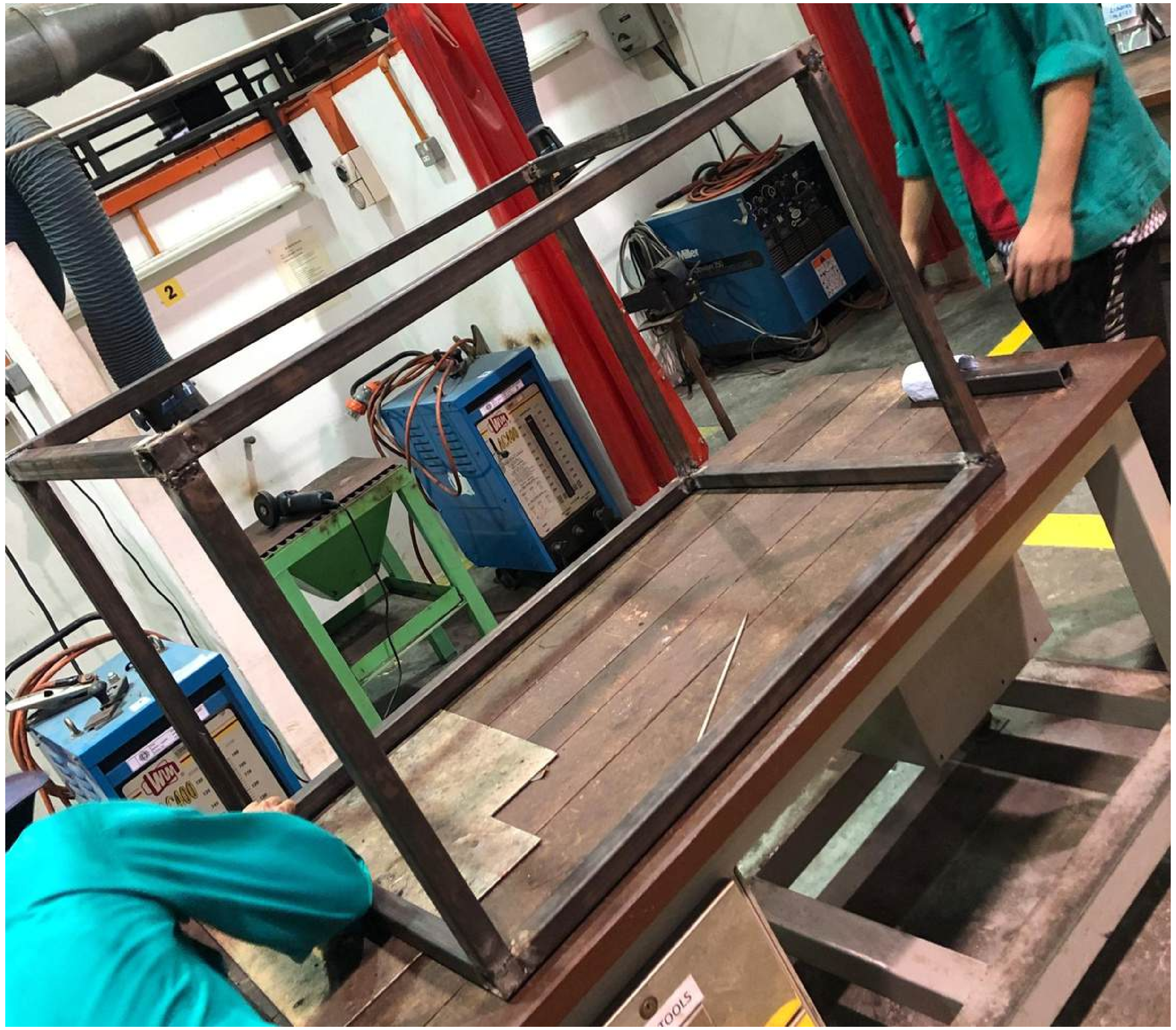


frame processing

2. Cutting & Welding Metal Components

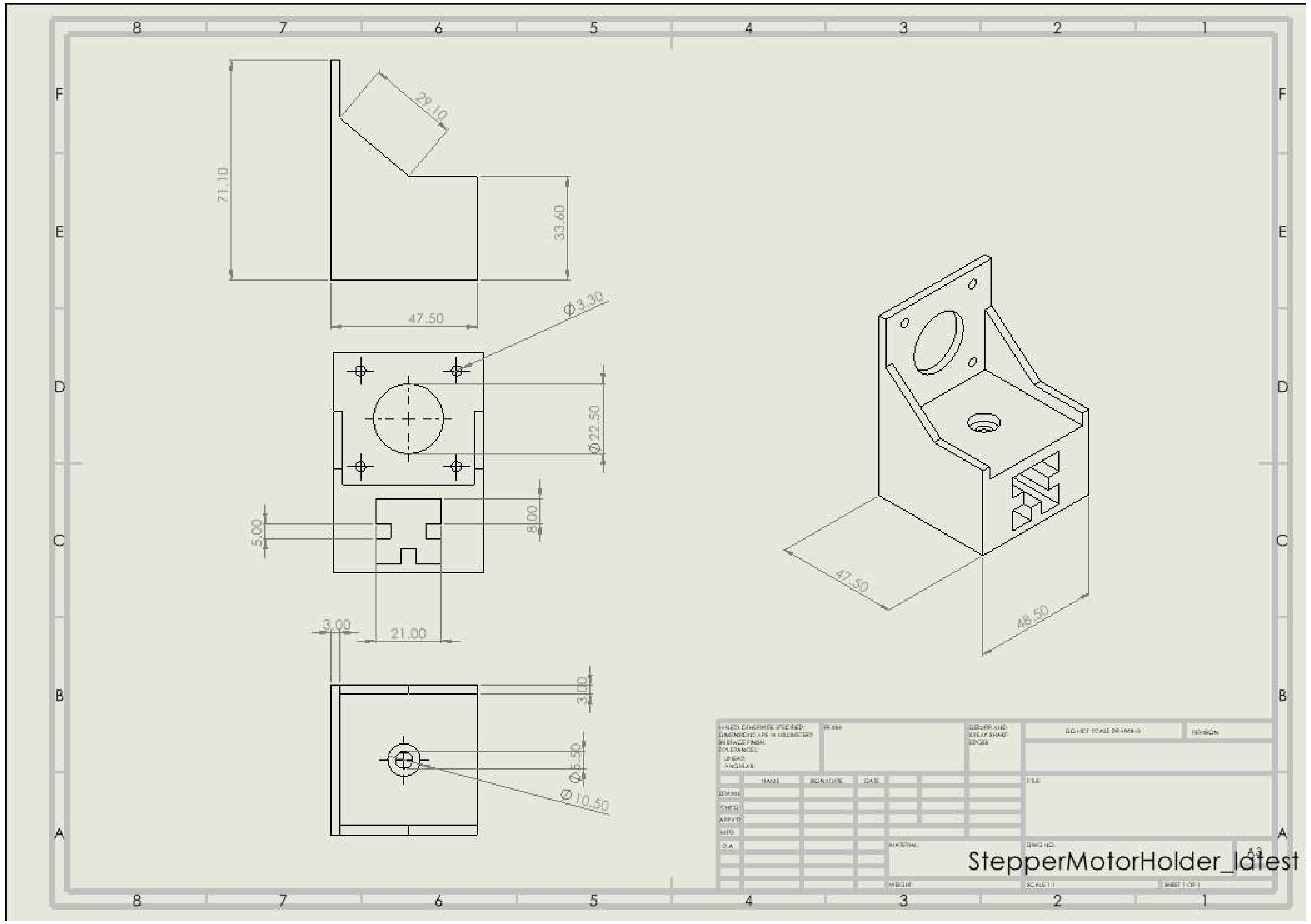
frame processing

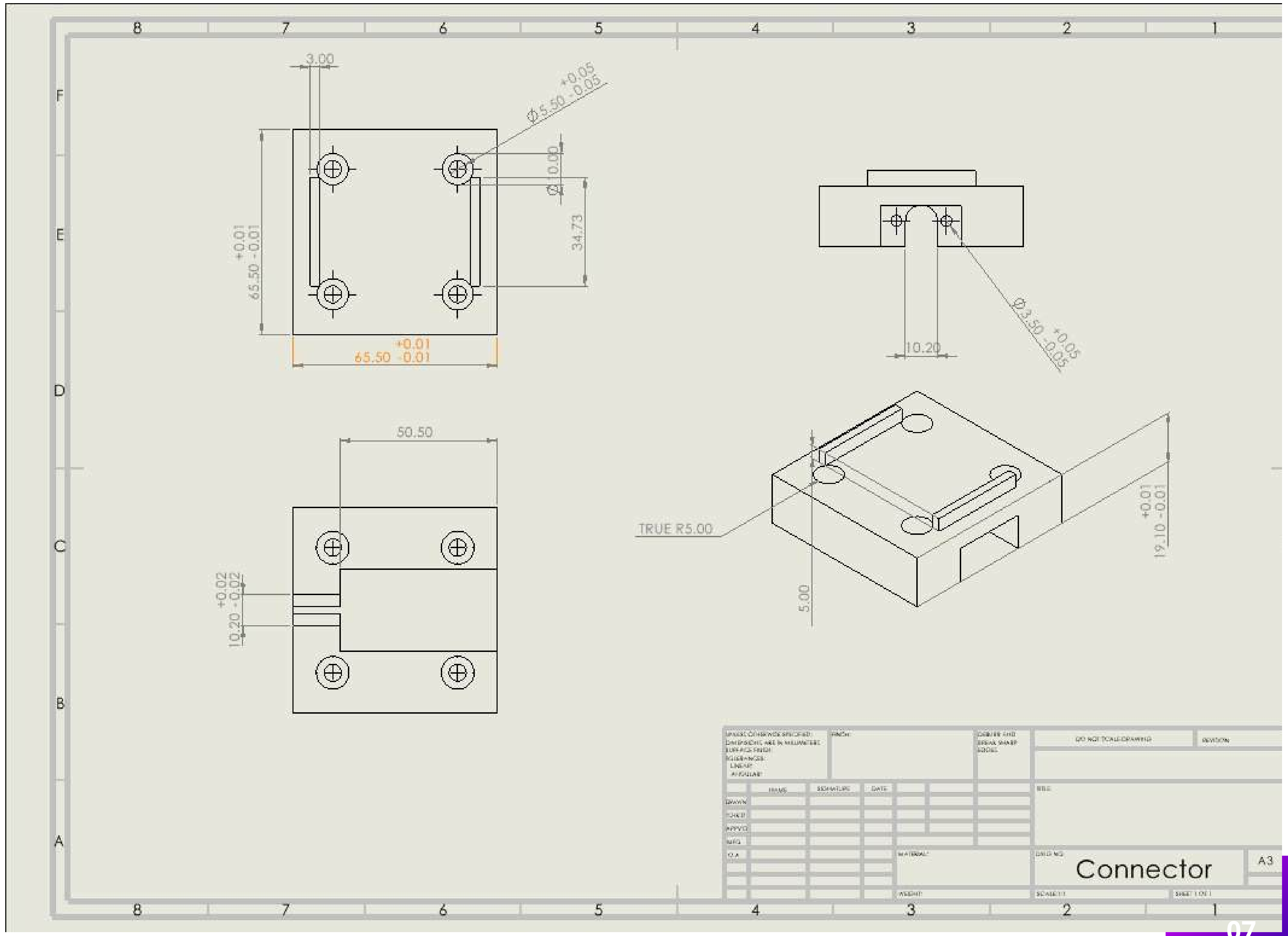
3. final frame shape



2. 3d printing Process

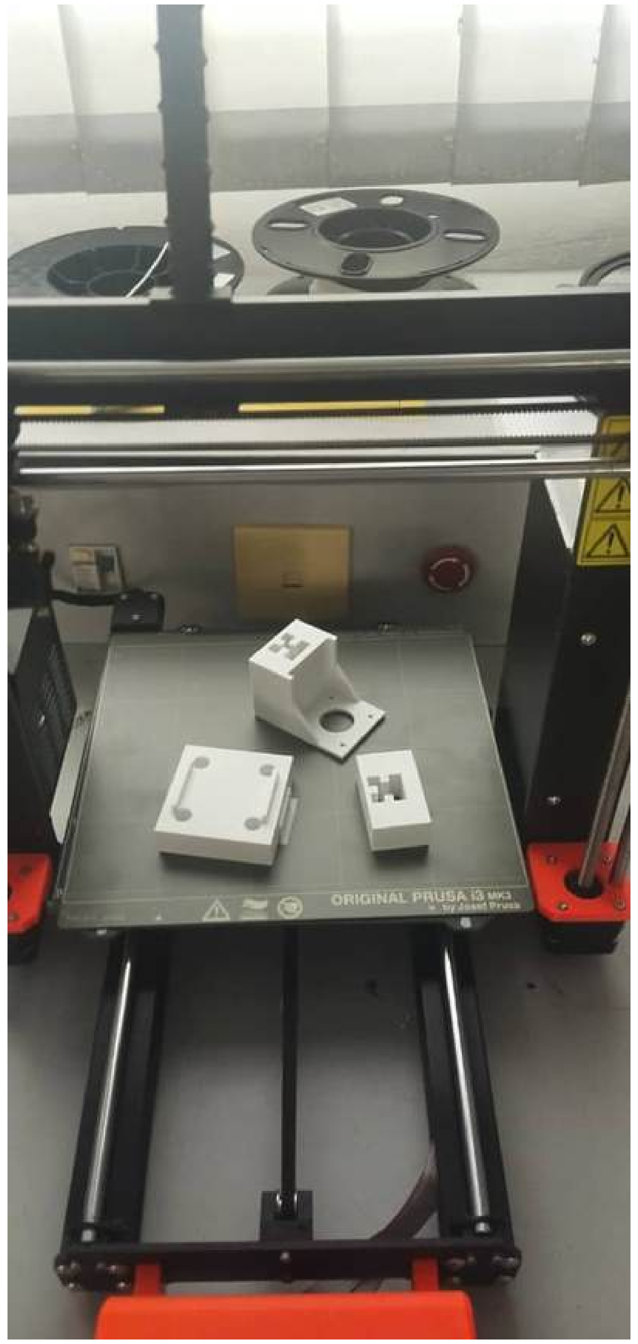
- supportings

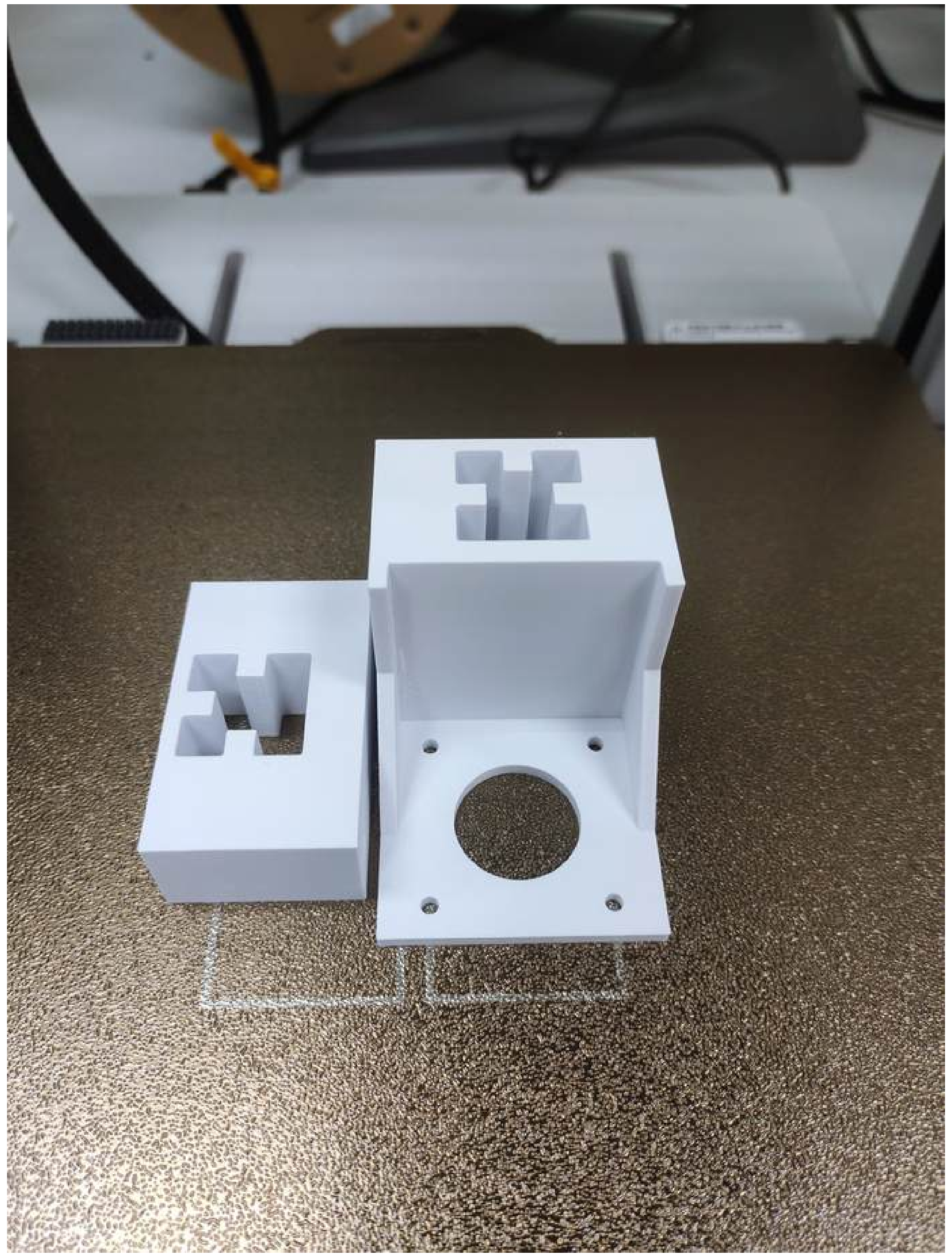




2. 3d printing Process

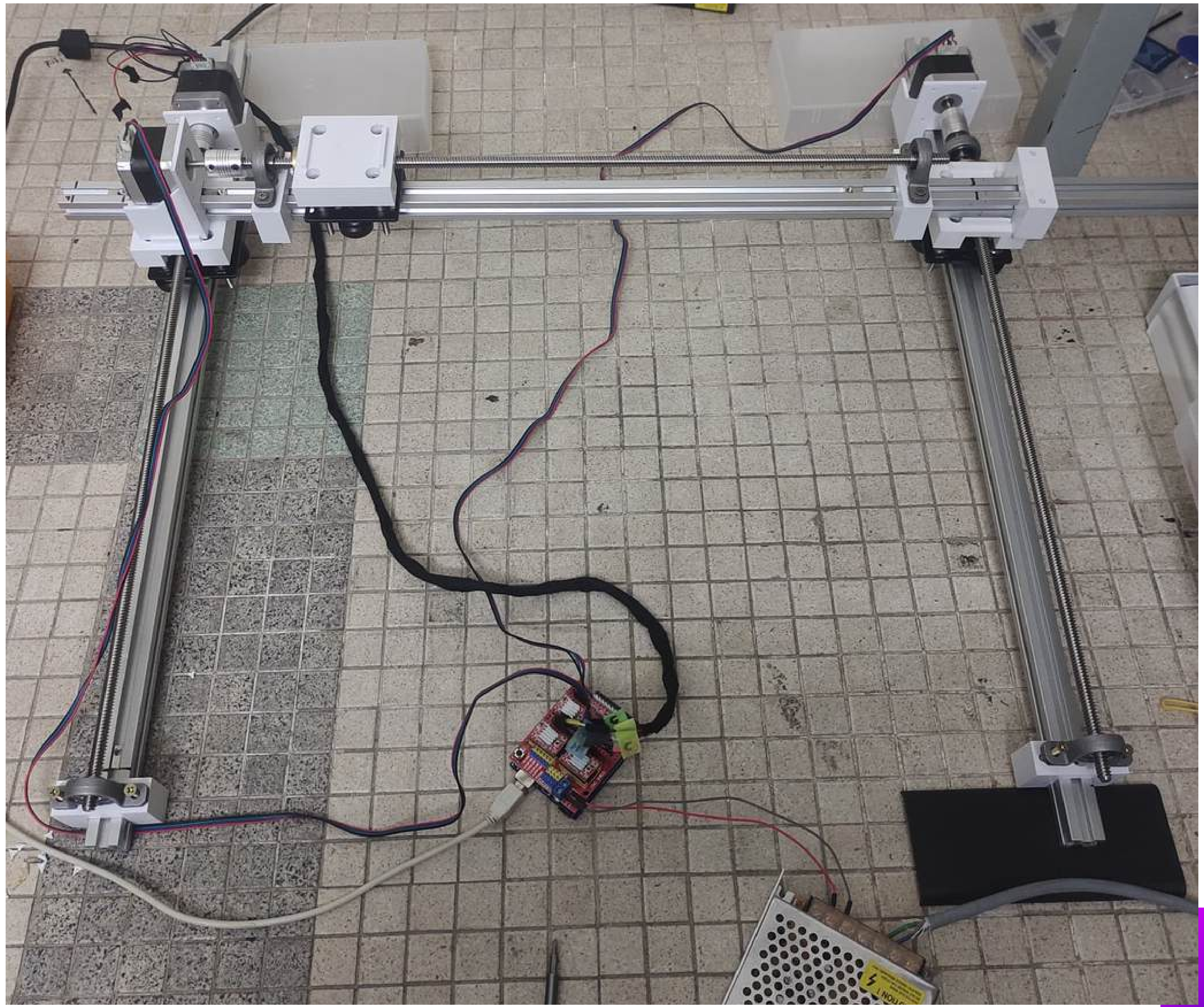
- Final design





3. Segregation Mechanism

1. Fabrication Process & Assembly Process



Segregation Mechanism

2. Testing the Gantry system Process

4. Calculation

1. Load and Torque Calculation

Key Considerations:

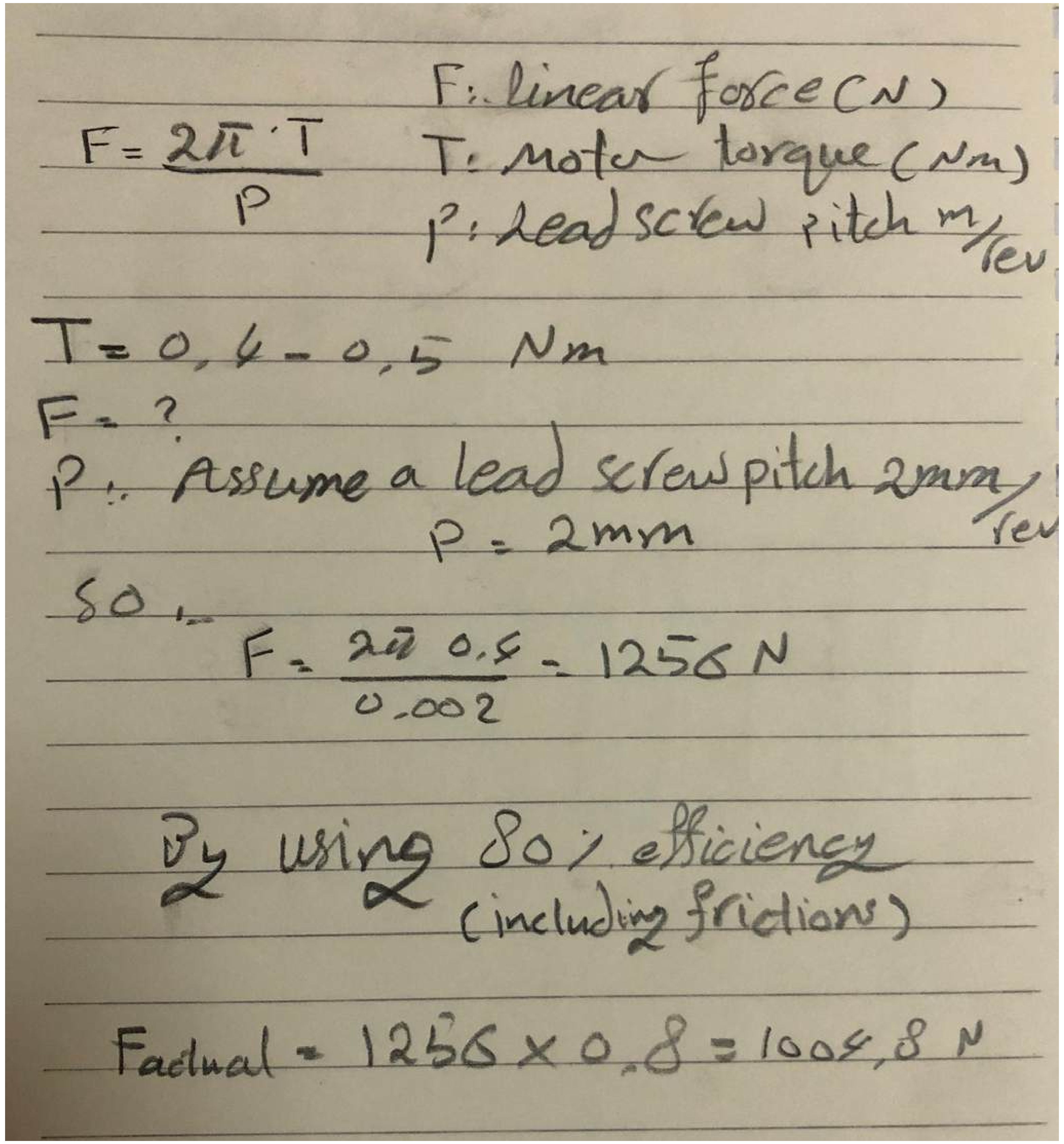
1. Motor Specifications:
   - Torque rating of the NEMA 17 stepper motors (typically 0.4–0.5 Nm for standard models at rated current).
   - Current and voltage supplied by the driver.
2. Mechanical Characteristics:
   - Lead screw pitch or belt pitch (linear movement per motor revolution).
   - Friction and efficiency of the guide rails and transmission system.
3. System Configuration:
   - Total load supported by Motor C (including its own weight, load, and frame).
   - Torque distribution on Motors A and B if they share the load.
4. Load Movement Steps:
   - Total steps to cover 5000 steps (as specified).

Calculation

- **Maximum focce**

- **Using the relationship between torque and linear force on the leadscrew.**

- **The theoretical maximum force is applied by each motor before friction and efficiency losses.**



$$F = \frac{2\pi \cdot T}{P}$$

F: linear force (N)
T: motor torque (Nm)
P: lead screw pitch m/rev

$T = 0,4 - 0,5$ Nm
$F = ?$
P: Assume a lead screw pitch 2mm/rev
$P = 2mm$

So,

$$F = \frac{2\pi \; 0,4}{0,002} = 1256 \text{ N}$$

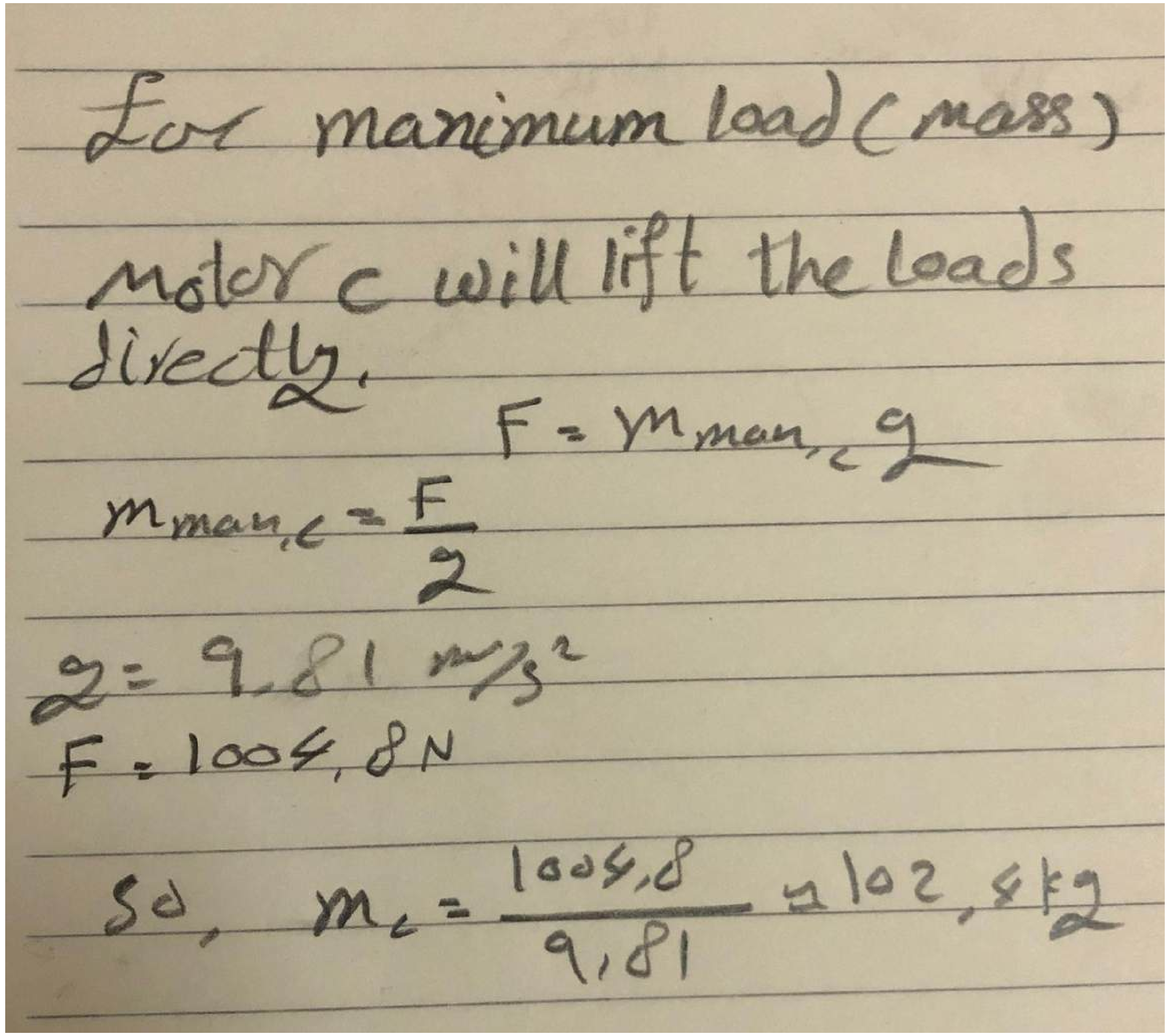
By using 80% efficiency (including frictions)

$F_{actual} = 1256 \times 0,8 = 1004,8$ N

Calculation

- **Maximum mass**

- **If Motor C carries a smaller load (e.g., less than 102.4 kg), the total capacity increases slightly.**



For maximum load (mass)

Motor c will lift the loads directly.

$$F = m_{max,c}\, g$$

$$m_{max,c} = \frac{F}{g}$$

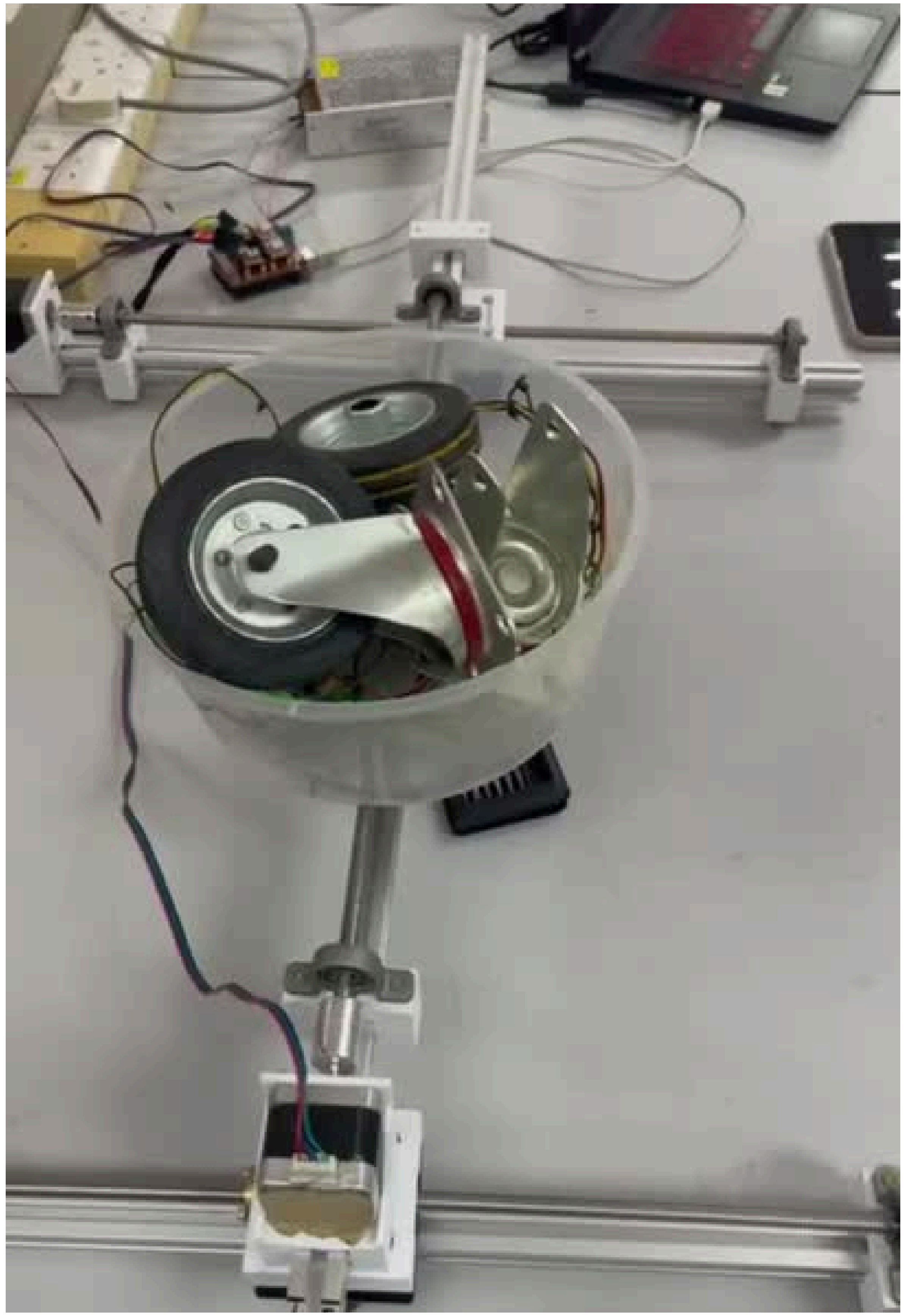
$$g = 9{,}81\ m/s^2$$

$$F = 1004{,}8\ N$$

$$\text{So,}\quad m_c = \frac{1004{,}8}{9{,}81} \approx 102{,}4\ kg$$

Calculation

- testing 6kg of mass

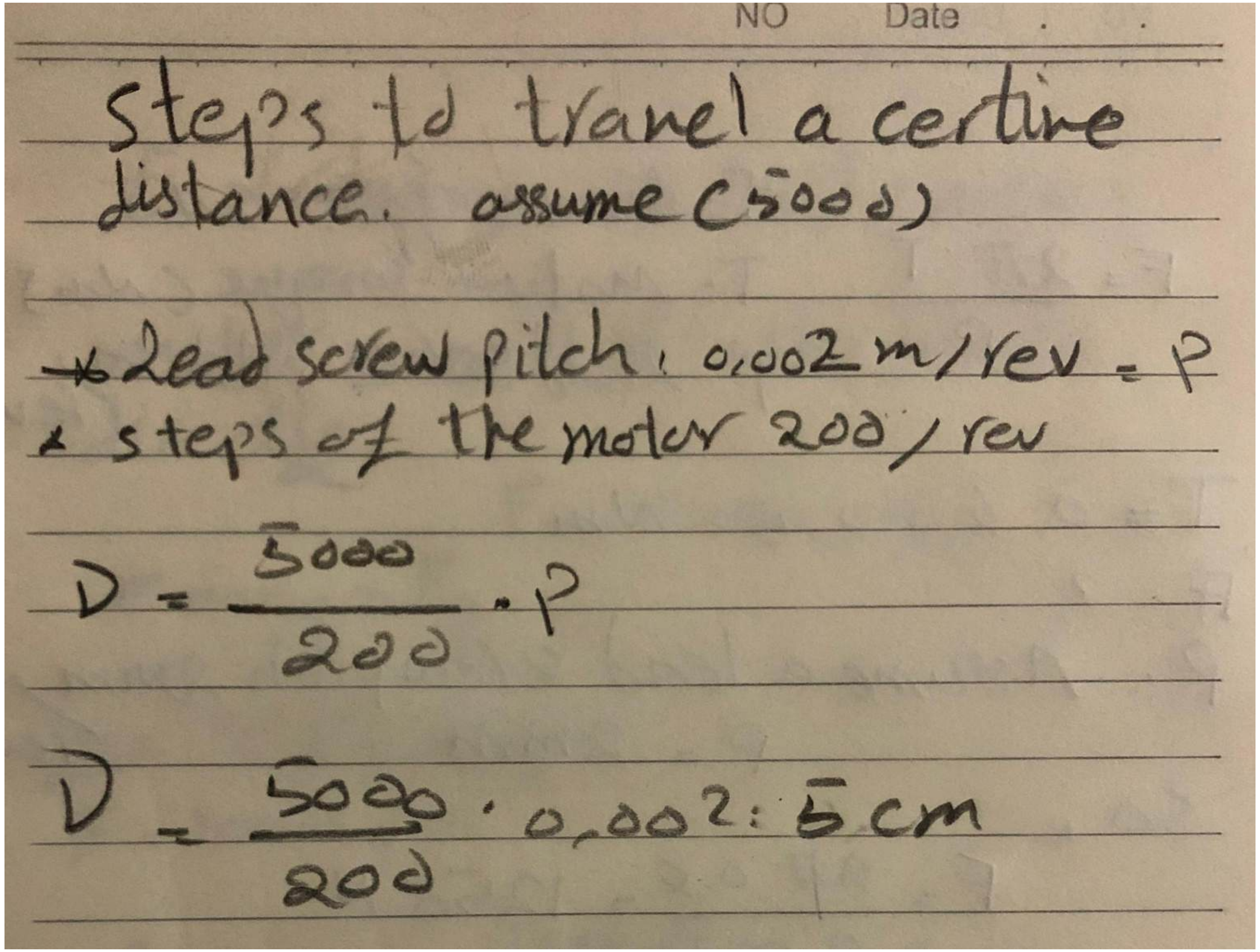


Calculation

- **distance calculation**

- **Steps for a Require Travelling**

- **assume (5000) steps**


NO Date

steps to travel a certine distance. assume (5000)

* lead screw pitch: 0,002 m/rev = P
* steps of the motor 200/rev

$$D = \frac{5000}{200} \cdot P$$

$$D = \frac{5000}{200} \cdot 0,002 = 5 cm$$

Machine Learning

Image Classification

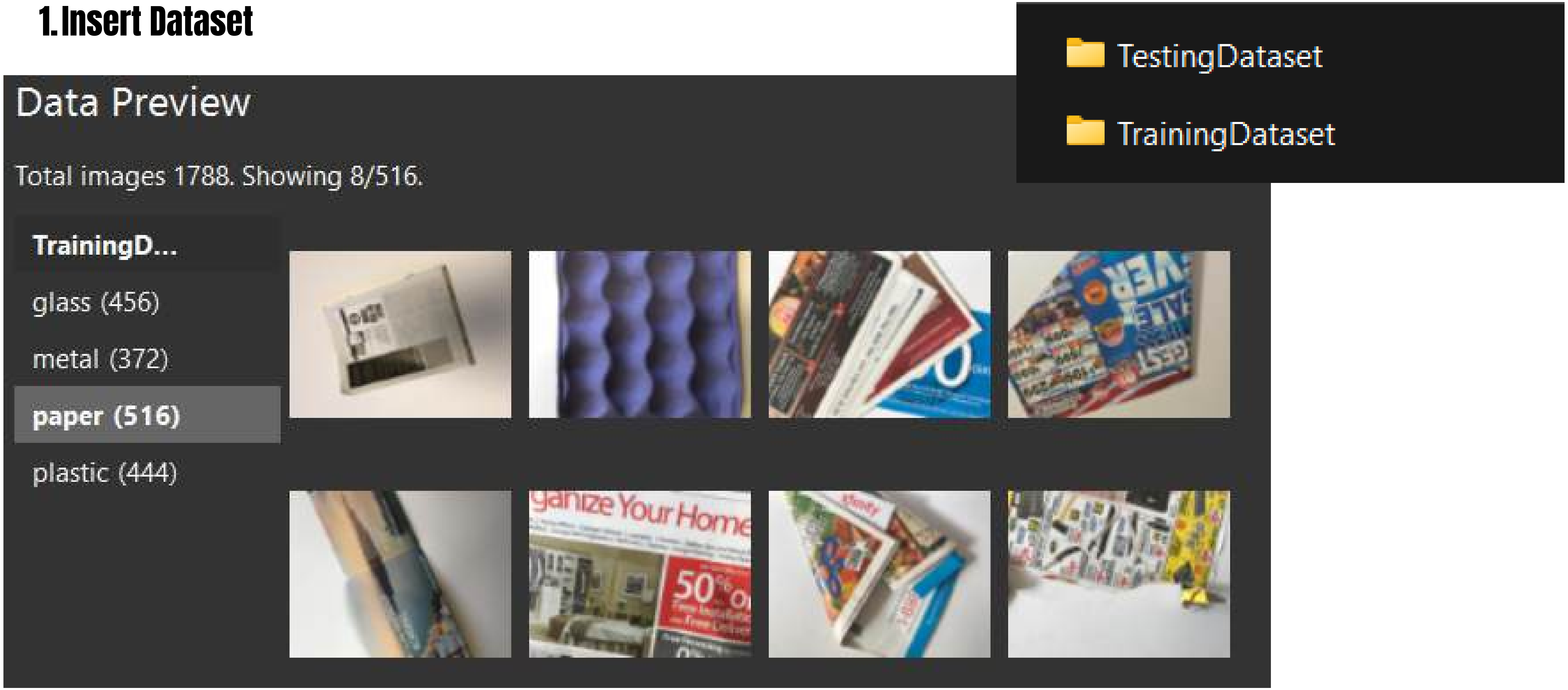
- Using ML.NET in Visual Studio 2022 for Image Classification

FLOW

Mechanical Parts → Preparing data set → Train Data Set → Evaluate Performance

Machine Learning

1. Insert Dataset



Machine Learning

2. Result by using this ML.Net Framework

Training results

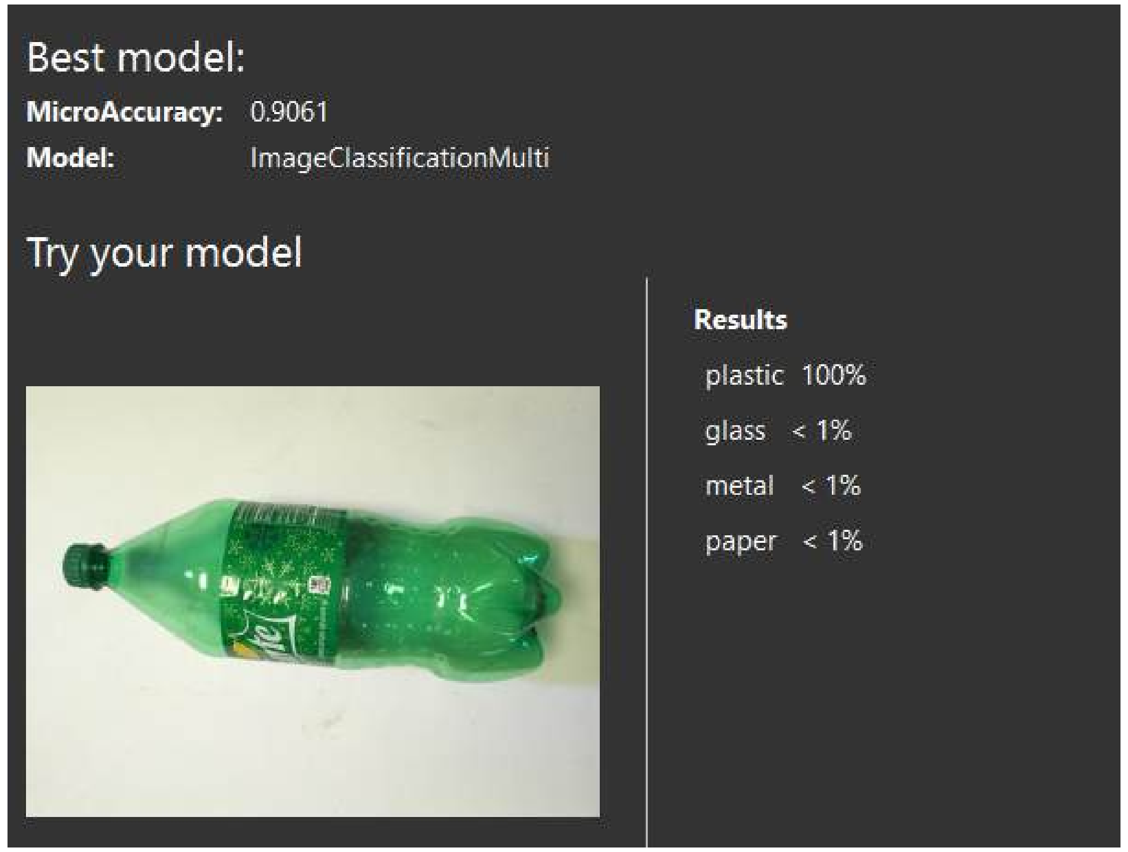
Best MicroAccuracy: 0.9061

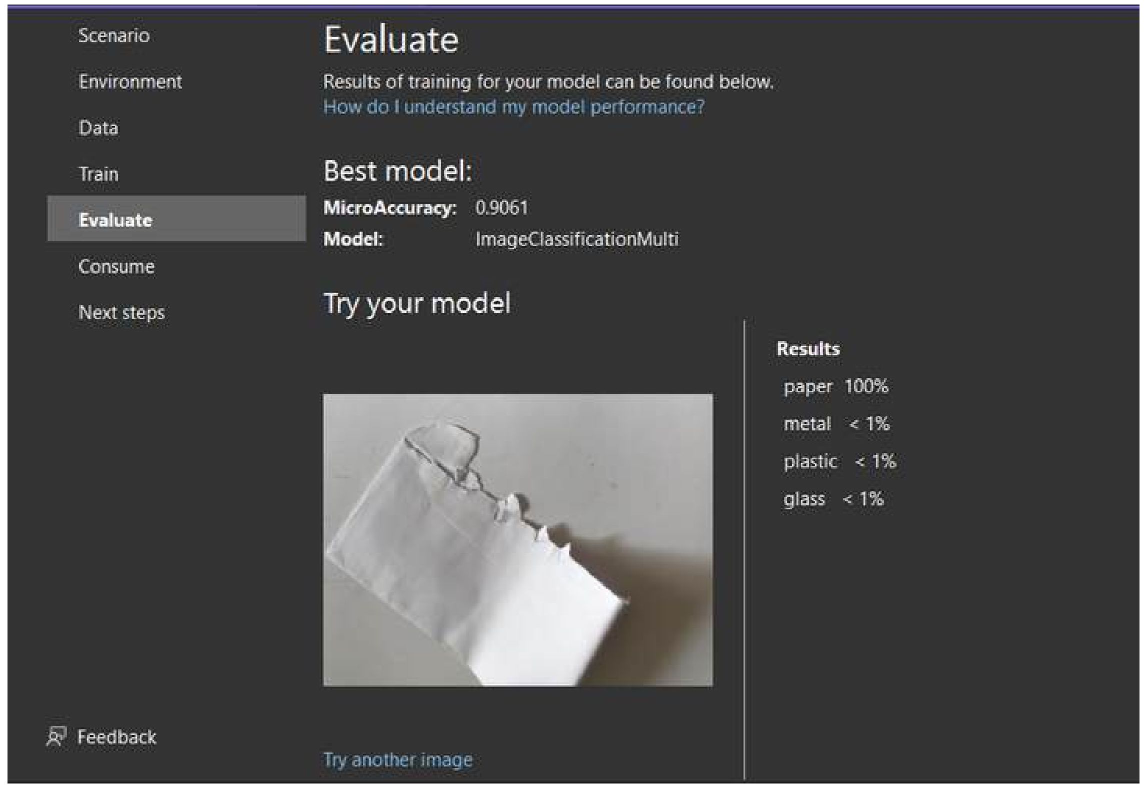
Best model: ImageClassificationMulti

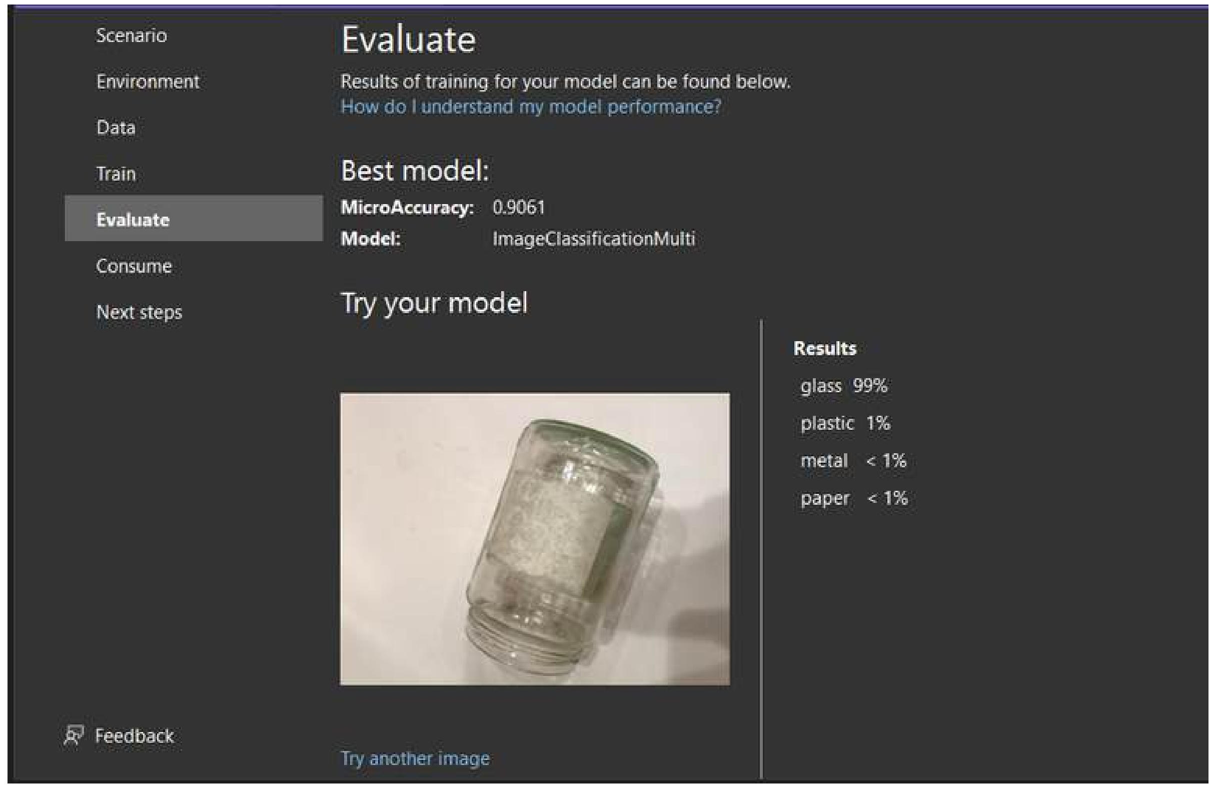
Training time: 285.18 seconds

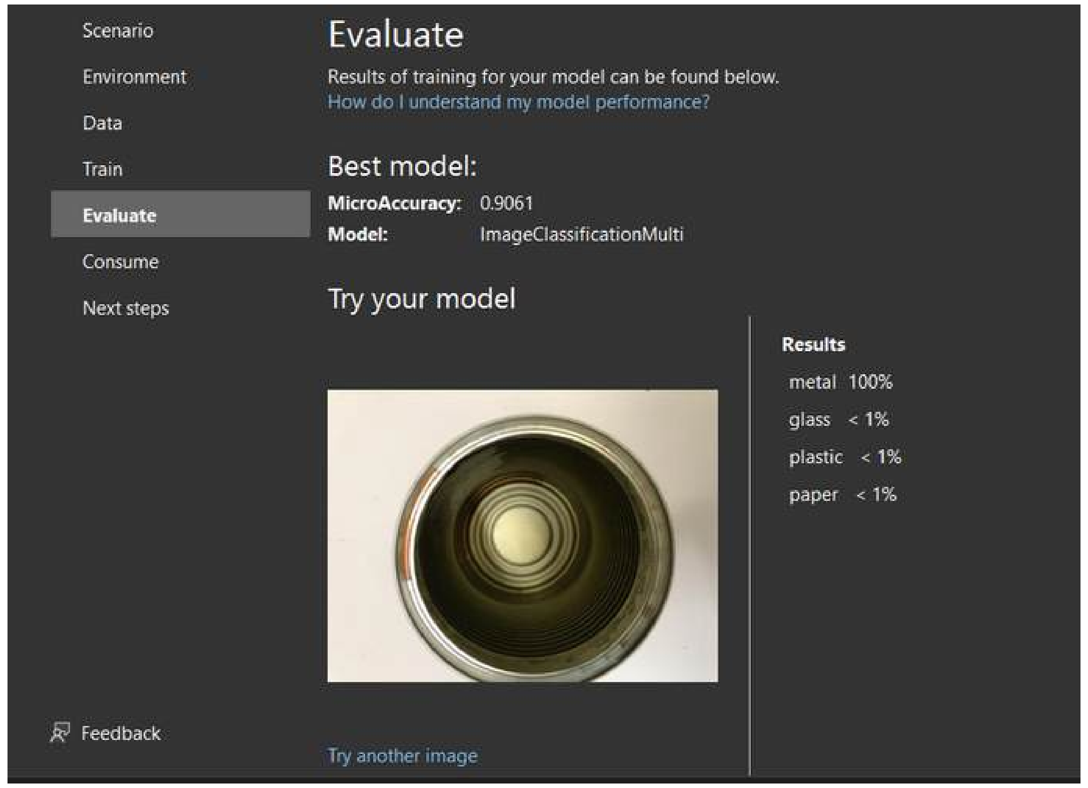
Models explored (total): 1

Generated code-behind: MLModel1.consumption.cs, MLModel1.training.cs









Machine Learning

Best model:
MicroAccuracy: 0.9061
Model: ImageClassificationMulti

Try your model

Results
- plastic 100%
- glass < 1%
- metal < 1%
- paper < 1%

Best model:
MicroAccuracy: 0.9061
Model: ImageClassificationMulti

Try your model

Results
- plastic 90%
- glass 10%
- metal < 1%
- paper < 1%

Best model:
MicroAccuracy: 0.9061
Model: ImageClassificationMulti

Try your model

Results
- metal 100%
- plastic < 1%
- glass < 1%
- paper < 1%

Best model:
MicroAccuracy: 0.9061
Model: ImageClassificationMulti

Try your model

Results
- metal 100%
- plastic < 1%
- paper < 1%
- glass < 1%

Best model:
MicroAccuracy: 0.9061
Model: ImageClassificationMulti

Try your model

Results
- paper 100%
- plastic < 1%
- metal < 1%
- glass < 1%

Best model:
MicroAccuracy: 0.9061
Model: ImageClassificationMulti

Try your model

Results
- paper 96%
- glass 3%
- metal 2%
- plastic < 1%

Best model:
MicroAccuracy: 0.9061
Model: ImageClassificationMulti

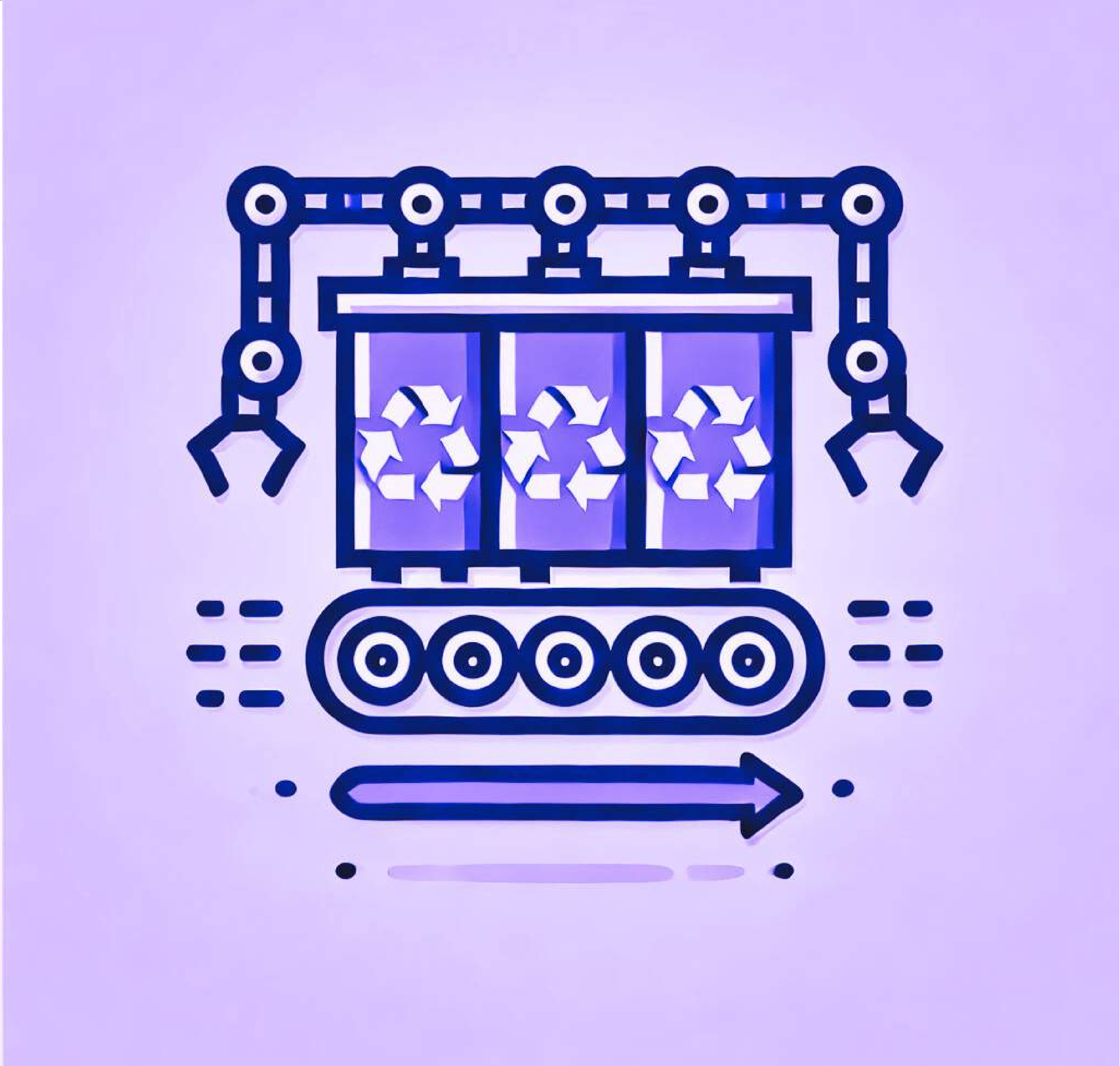
Try your model

Results
- plastic 94%
- paper 6%
- glass 1%
- metal < 1%

Best model:
MicroAccuracy: 0.9061
Model: ImageClassificationMulti

Try your model

Results
- glass 100%
- metal < 1%
- plastic < 1%
- paper < 1%



THANK YOU